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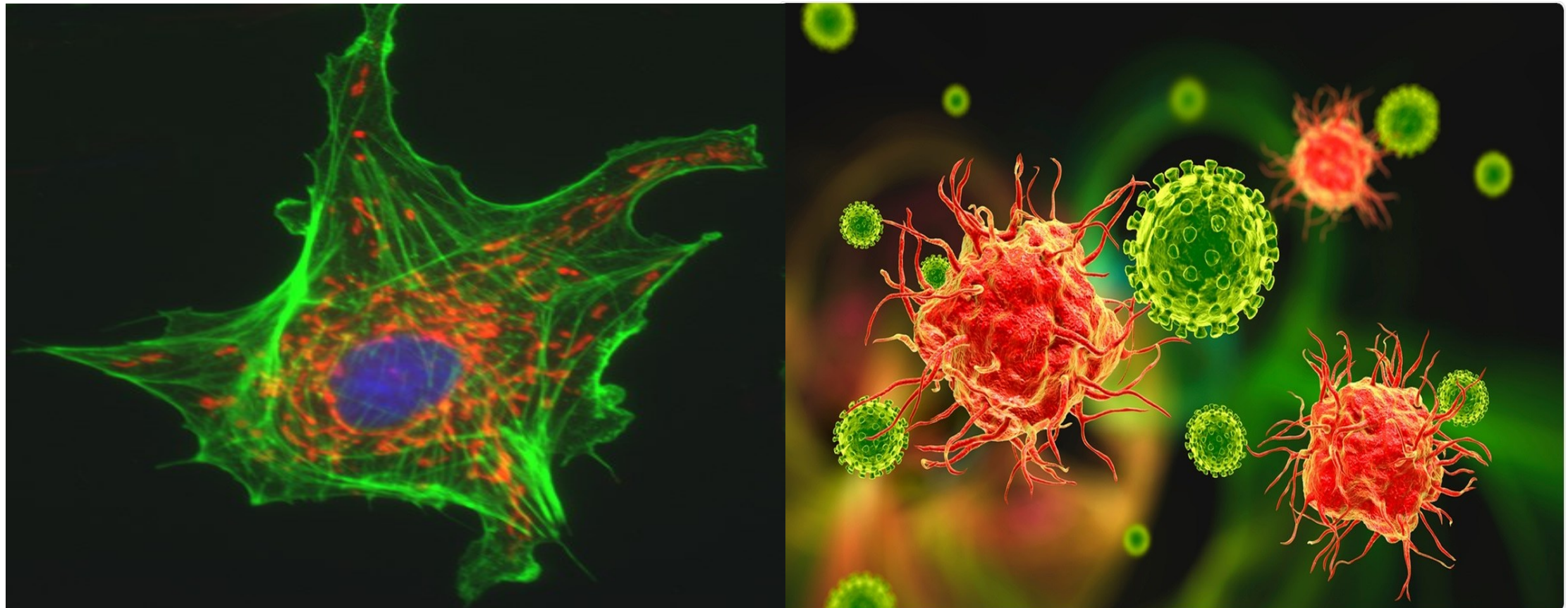
FIELD OF SPECILIZATION: BIOMEDICAL SCIENCE & TOXICOLOGY

***RESEARCH AREA: GENETIC TOXICOLOGY, MECHANISTIC TOXICOLOGY, ENVIRONMENTAL TOXICOLOGY,
NANOTOXICOLOGY, MICROBIOLOGY, CELL CULTURE & MODE OF CELL DEATH***

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THE CELL: Part 1

Siew Ee Ling, PhD



Leave NO ONE behind!!!



ICE BREAKING!

If you were a cell, what type of cell would you be and why?

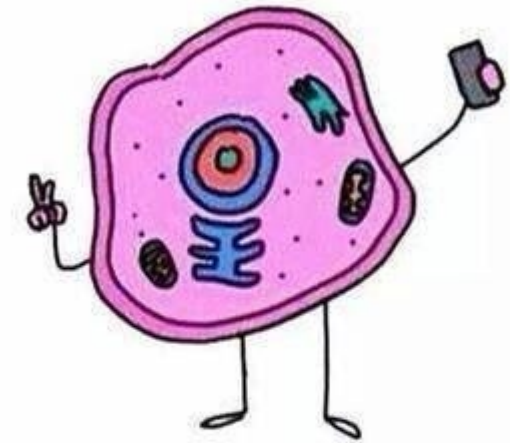
- Think creatively about different cell types
- Consider the functions and characteristics of various cells
- Relate cell biology to their own personalities or interests

A neuron because they love to communicate

A muscle cell because they're athletic

A white blood cell because they enjoy helping others

A plant cell because they love being outdoors



Cell-fie

Cell specialization, the diversity of cell types, and how cell structure relates to function.

Ice Breaking: What cell identity matches your personality?

A Neuron

"Because I love to communicate."

- Highly specialized
- Sends signals in networks
- Coordinates systems
- Loves sharing ideas

A Muscle Cell

"Because I am athletic & active."

- Strong and resilient
- Packed with mitochondria
- Works in absolute sync
- Highly energetic

A White Blood Cell

"Because I enjoy helping others."

- Protective and helpful
- Patrols for invaders
- Performs phagocytosis
- Defends the system

A Plant Cell

"Because I love being outdoors."

- Structured and green
- Resilient cell walls
- Performs photosynthesis
- Sustains life on Earth

Why Study Cells? The universal unit of biological structure

All biological activities originate at the microscopic scale. By examining cells, we observe how life operates at the threshold.

- 1. Universal Foundation:** Cells are the absolute smallest units that perform all 8 essential functions of living organisms.
- 2. Structure Dictates Function:** Specialized cell shapes—from nerve impulses to protective tissue structures—allow organisms to survive.
- 3. Health & Mechanistic Science:** Diseases, cell death mechanisms, and nanotoxicity originate as sub-cellular failures, making cytology vital for medicine .

LECTURE INQUIRY & CRITICAL THINKING

'If cells are the fundamental units of life, why can a single amoeba live as an independent organism, while a single human skin cell cannot survive in isolation?'

What Is a Cell? The Three Tenets of Cell Theory

Tenet 01

All living things are composed of cells

- Every single biological organism—from unicellular prokaryotes to multicellular oaks—is constructed from cells.
- Cells are the ultimate physical building blocks of life.

Tenet 02

Cells are the basic unit of life

- The cell represents the smallest physical entity capable of maintaining biological order and executing metabolism.
- Cell structure is closely linked to its specialized roles.

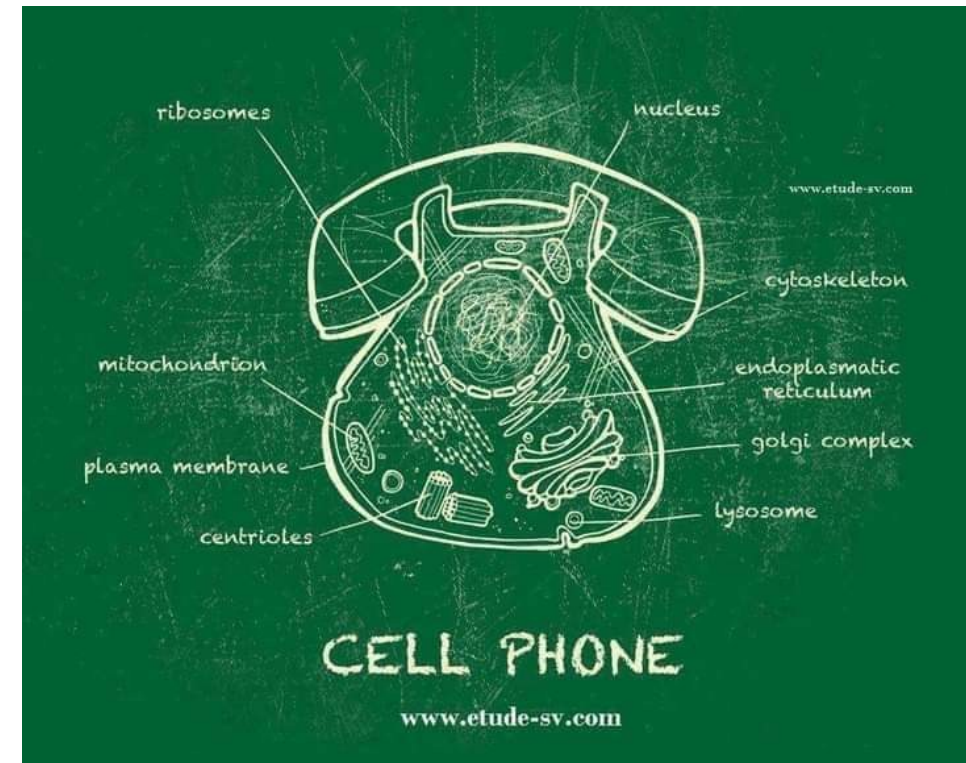
Tenet 03

All cells arise from pre-existing cells

- Cells do not generate spontaneously; they arise only via physical cell division of mother cells.
- This continuity ensures the preservation of hereditary DNA.

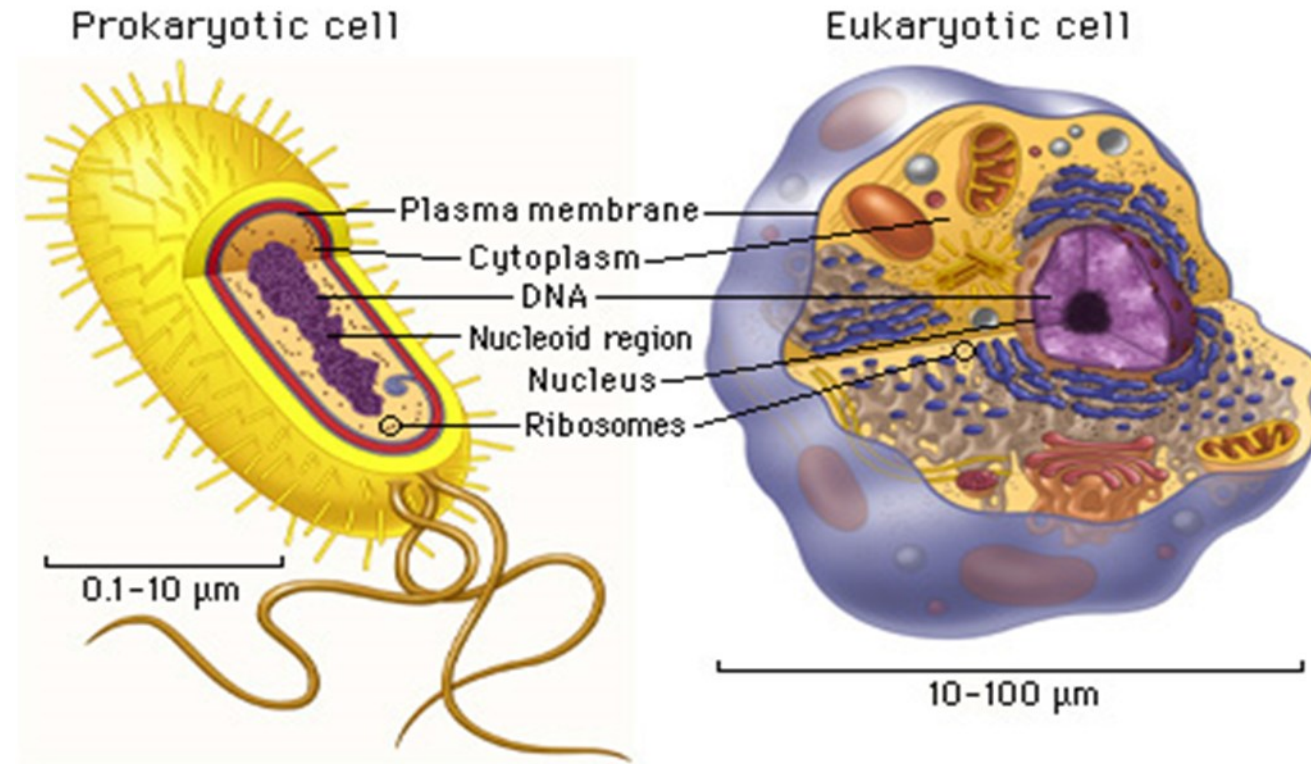
Cell Theory

- All living things are composed of cells
- Cells are the basic unit of all organisms
- All cells arise from preexisting cells



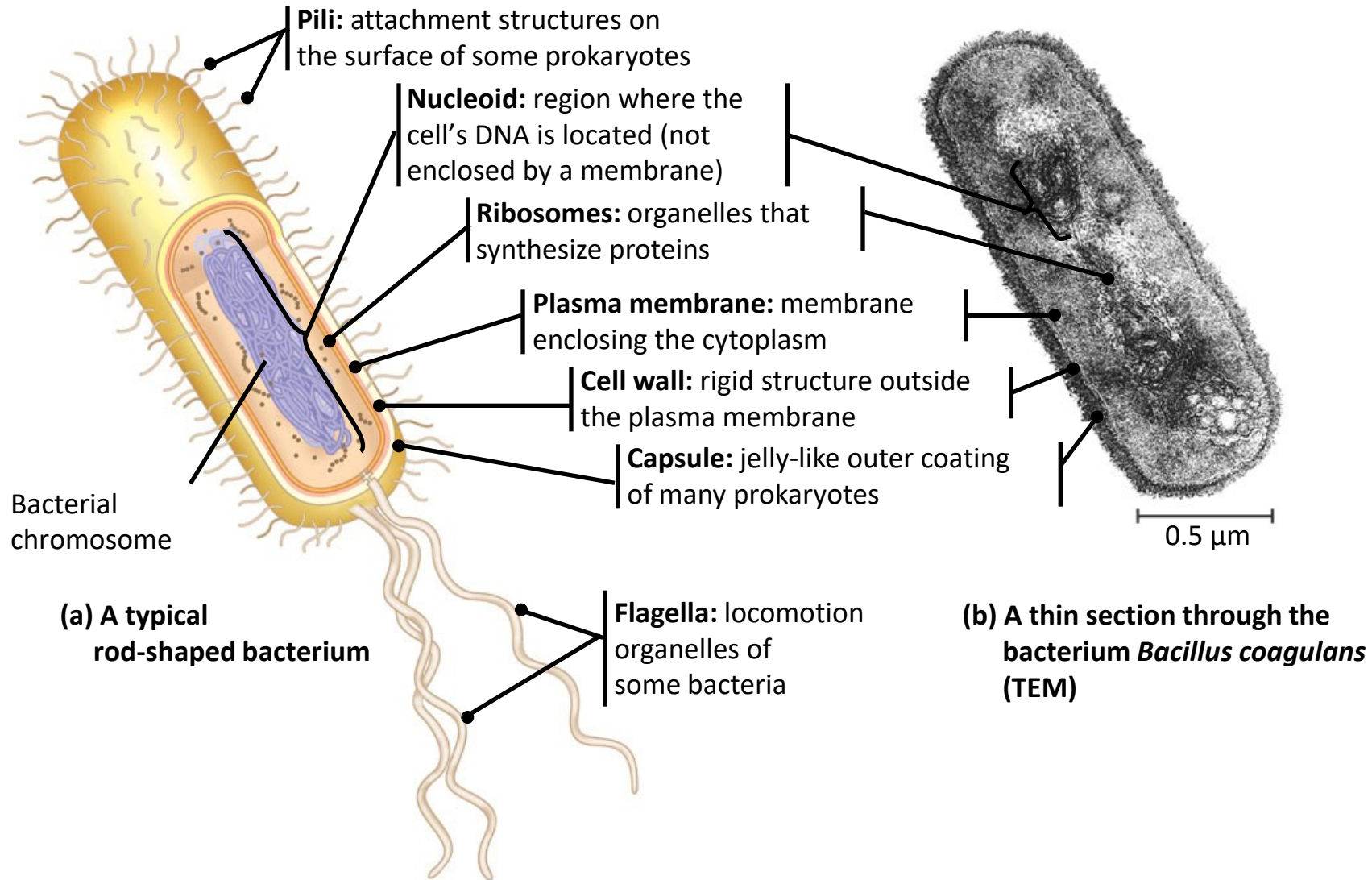
Types Of Cells

- Prokaryotic cell (prokaryote) & Eukaryotic cell (eukaryote)
- Prokaryotes- cells **without** true nucleus or other internal membrane
- Eukaryotes – cells **with** true nucleus and organelles surrounded by a membrane.



Prokaryote

- It can be photoautotroph, chemoautotroph, photoheterotroph, chemoheterotroph
- **Photoautotroph** – get E from light & C from fixing CO₂
- **Chemoautotroph** – get E chemically & C from fixing CO₂
- **Photoheterotroph** – get E from light, receive C from ingesting other organisms
- **Chemoheterotroph** – get E & C from organic compounds by ingesting organisms @ materials with C



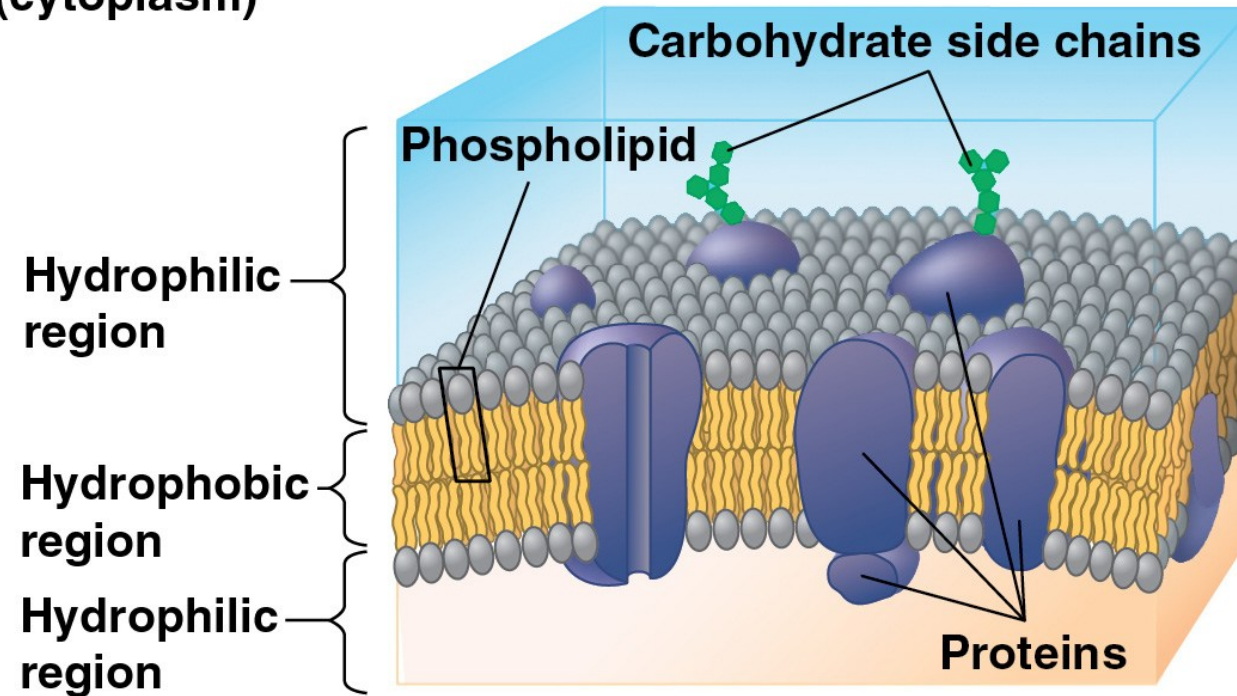
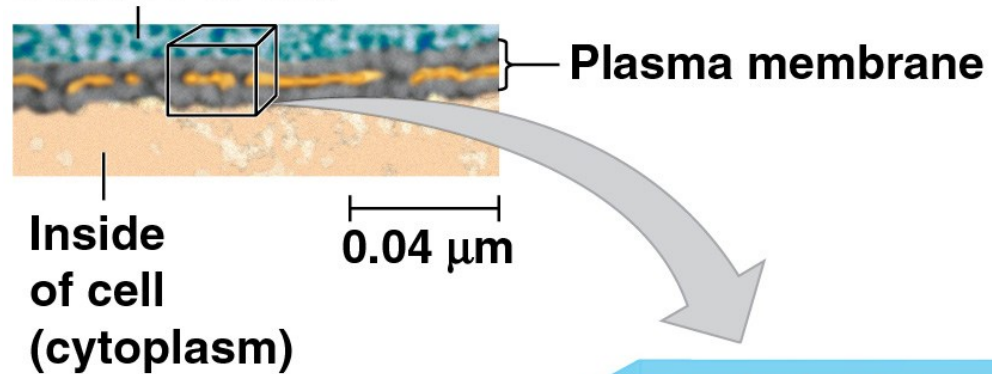
Eukaryote

- **Eukaryotic cells** are characterized by having
 - DNA in a nucleus that is bounded by a double membrane
 - Membrane-bound organelles
 - **Cytoplasm** in the region between the plasma membrane and nucleus
- Eukaryotic cells are generally much larger than prokaryotic cells

- Metabolic requirements set upper limits on the size of cells
- The **plasma membrane** is a selective barrier that allows sufficient passage of oxygen, nutrients, and waste to service the volume of every cell
- The surface area to volume ratio of a cell is critical
- As a cell increases in size, its volume grows proportionately more than its surface area

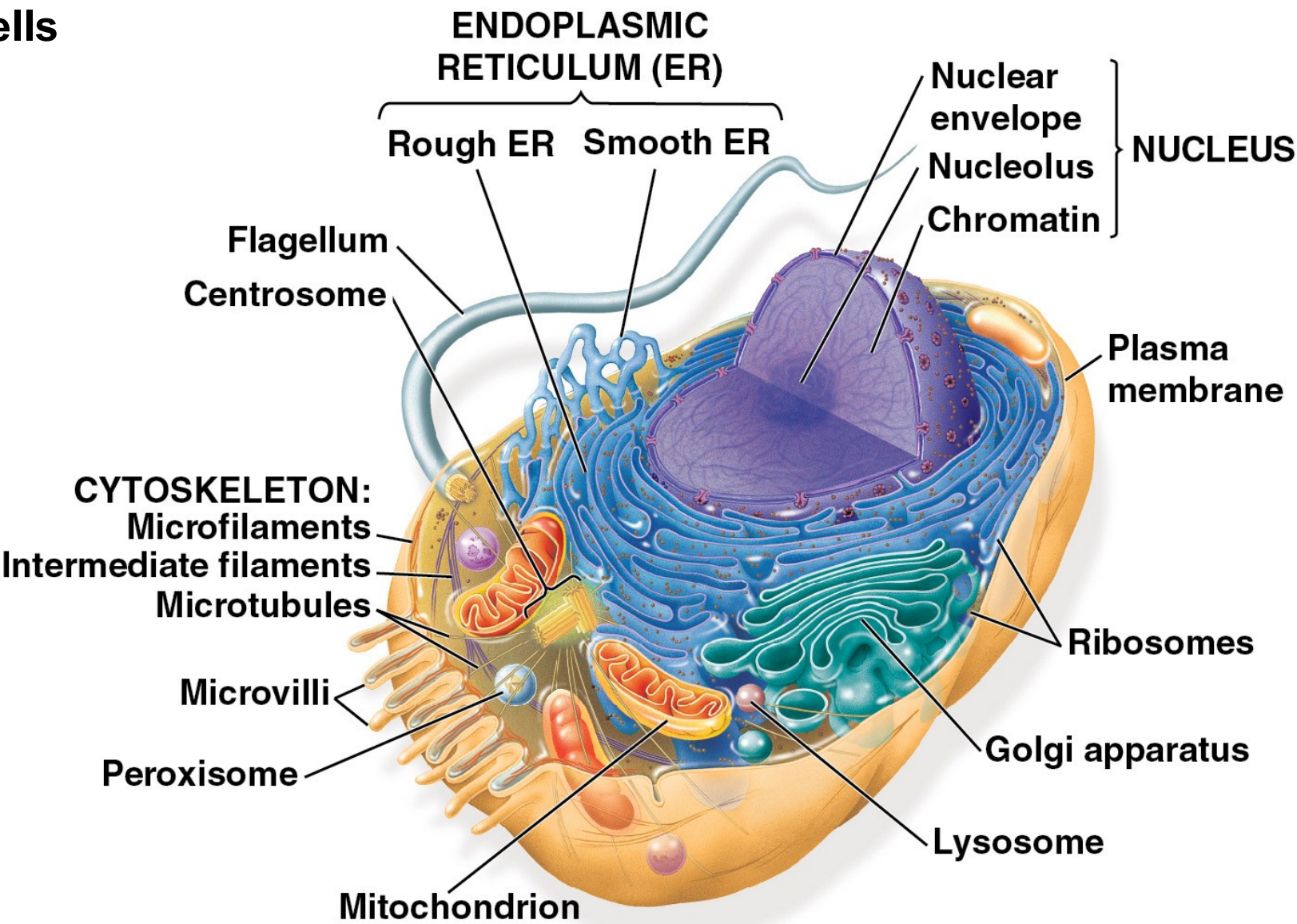
(a) Colorized TEM of a plasma membrane

Outside of cell



(b) Structure of the plasma membrane

Exploring eukaryotic cells



How does the internal organization of eukaryotic cells allow them to perform the functions of life?

Energy and matter transformations

A system of internal membranes synthesizes and modifies proteins, lipids, and carbohydrates.

Chloroplasts convert light energy to chemical energy.

Mitochondria break down molecules, generating ATP.

Interactions with the environment

The plasma membrane controls what goes into and out of the cell.

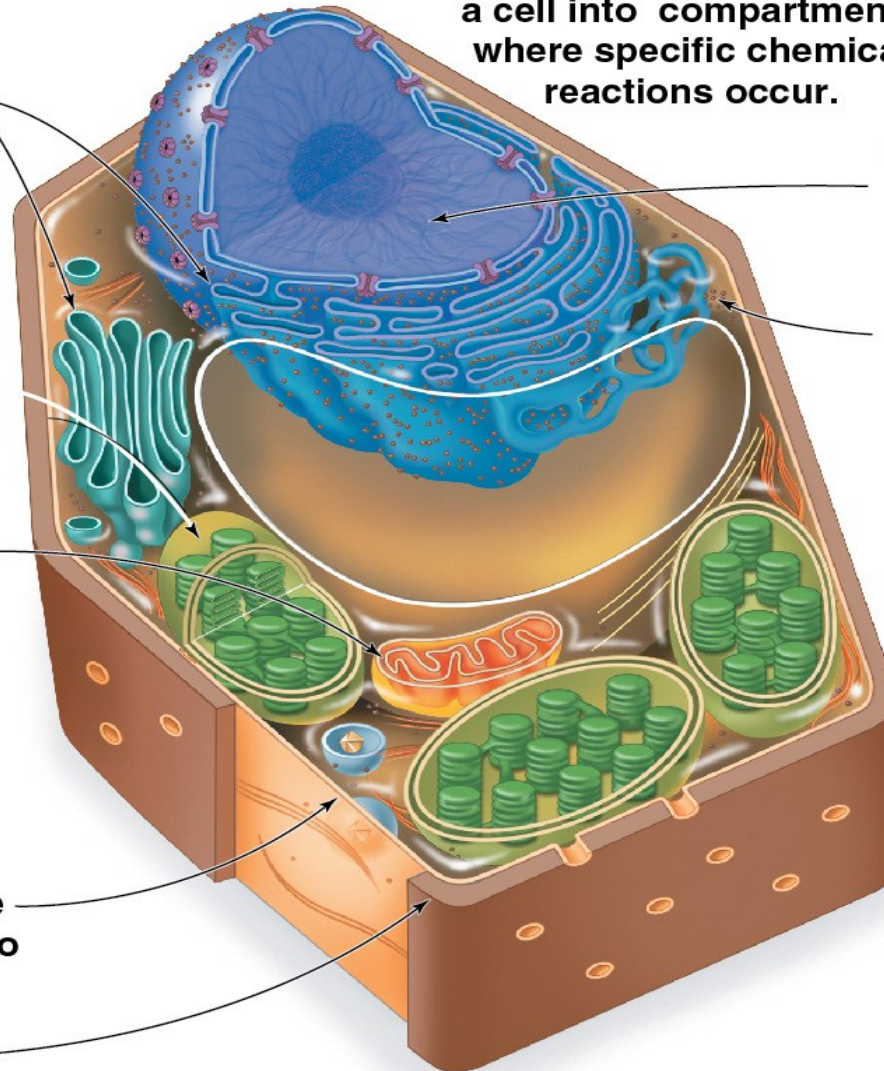
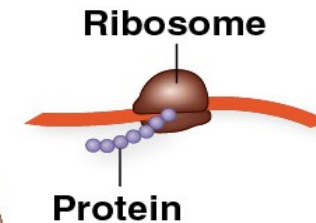
Plant cells have a protective cell wall.

Internal membranes divide a cell into compartments where specific chemical reactions occur.

Genetic information storage and transmission

DNA in the nucleus contains instructions for making proteins.

Ribosomes are the sites of protein synthesis.



PROKARYOTES vs EUKARYOTES

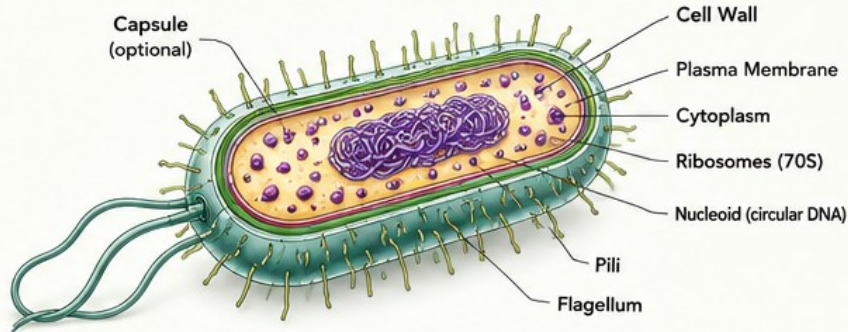
Two Major Types of Cells, One Diversity of Life



All living things are made of cells. **Prokaryotes** are simpler, usually single-celled organisms. **Eukaryotes** have more complex cells and include all plants, animals, fungi, and protists.

PROKARYOTES

Found in **Bacteria** and **Archaea**.



SIZE

Typically 0.1 – 5 μm



NUCLEUS

No true nucleus.
DNA is in the nucleoid region.



MEMBRANE-BOUND ORGANELLES

Absent



RIBOSOMES

70S (smaller)



CELL DIVISION

Binary fission



METABOLISM

Occurs on the plasma membrane
(e.g., respiration, photosynthesis in some)



ORGANISMS

Unicellular only



EXAMPLES

Escherichia coli (bacterium),
Methanobacterium (archaea)

FEATURE



SIZE



NUCLEUS



MEMBRANE-BOUND ORGANELLES



RIBOSOMES



CELL DIVISION



METABOLISM



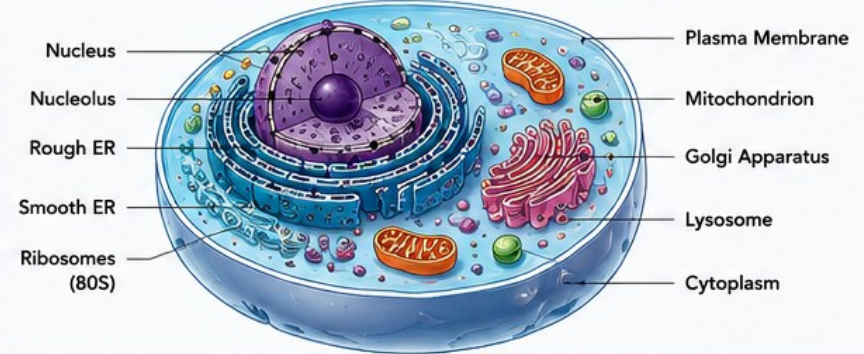
ORGANISMS



EXAMPLES

EUKARYOTES

Found in **Protists**, **Fungi**, **Plants**, and **Animals**.



SIZE

Typically 10 – 100 μm

NUCLEUS

True nucleus enclosed by nuclear envelope;
contains linear DNA with histone proteins.

MEMBRANE-BOUND ORGANELLES

Present (e.g., mitochondria, ER, Golgi, lysosomes,
chloroplasts in plants)

RIBOSOMES

80S (larger) in cytoplasm; 70S in mitochondria
and chloroplasts

CELL DIVISION

Mitosis (for growth) and meiosis (for reproduction)

METABOLISM

Occurs in specialized organelles
(e.g., mitochondria, chloroplasts)

ORGANISMS

Mostly multicellular; some unicellular
(e.g., yeast, amoeba)

EXAMPLES

Human cells, onion cells, yeast (fungus),
Amoeba (protist)



KEY TAKEAWAY

Prokaryotes are simple and efficient, thriving in diverse environments.
Eukaryotes are complex and compartmentalized, enabling specialization and multicellular life.



BACTERIUM



YEAST

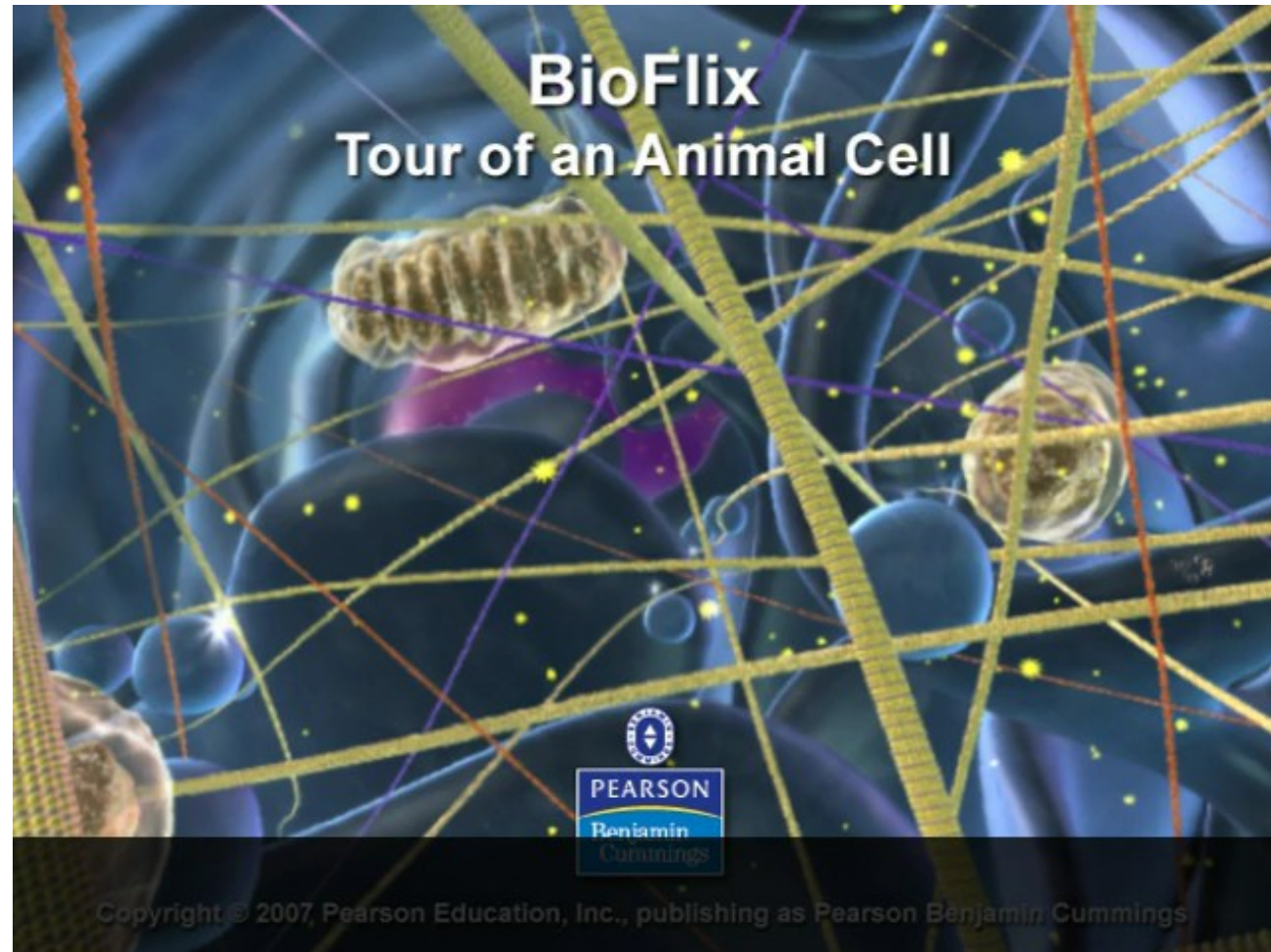


PLANT CELL



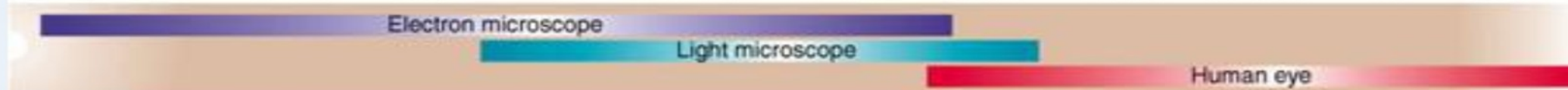
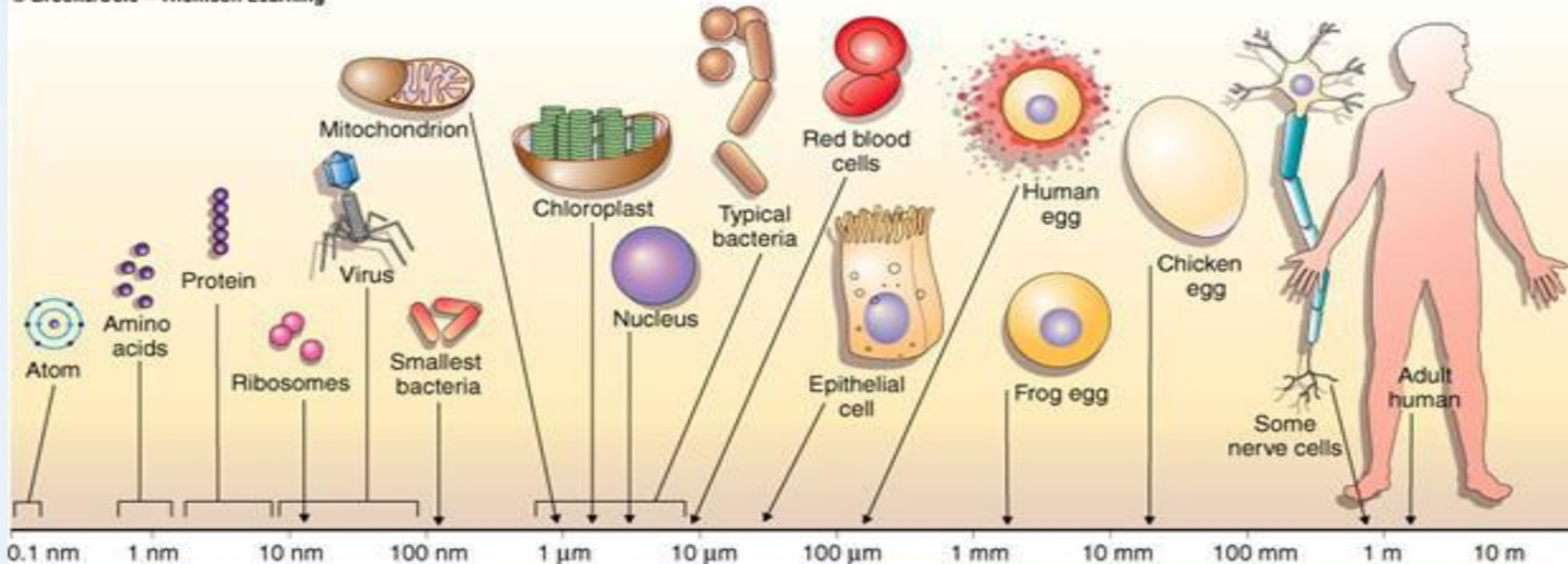
ANIMAL CELL

BioFlix® Animation: Tour of an Animal Cell



Biological size and cell diversity

© Brooks/Cole – Thomson Learning



Measurements

1 meter	=	1000 millimeters (mm)
1 millimeter	=	1000 micrometers (μ m)
1 micrometer	=	1000 nanometers (nm)

CHECKPOINT!

Explain the functional significance of the cell size and cell shape.



Critical factor in determining cell size is the ratio of plasma membrane (surface area) to the cell's volume. The plasma membrane must be large enough relative to the cell volume to regulate the passage of materials in and out the cells

How Scientists Study Cells????

Robert Hooke: identified the cell walls and gave the structure the name “cell”. Study of the cell was not possible until microscopes.



Methods for Studying Cells

- To study cells, biologists use microscopes and the tools of biochemistry
- Scientists use microscopes to visualize cells too small to see with the naked eye

Microscopy

- Microscopes are used to visualize cells
- In **a light microscope (LM)**, visible light is passed through a specimen and then through glass lenses
- Lenses refract (bend) the light so that the image is magnified

Microscope Parts

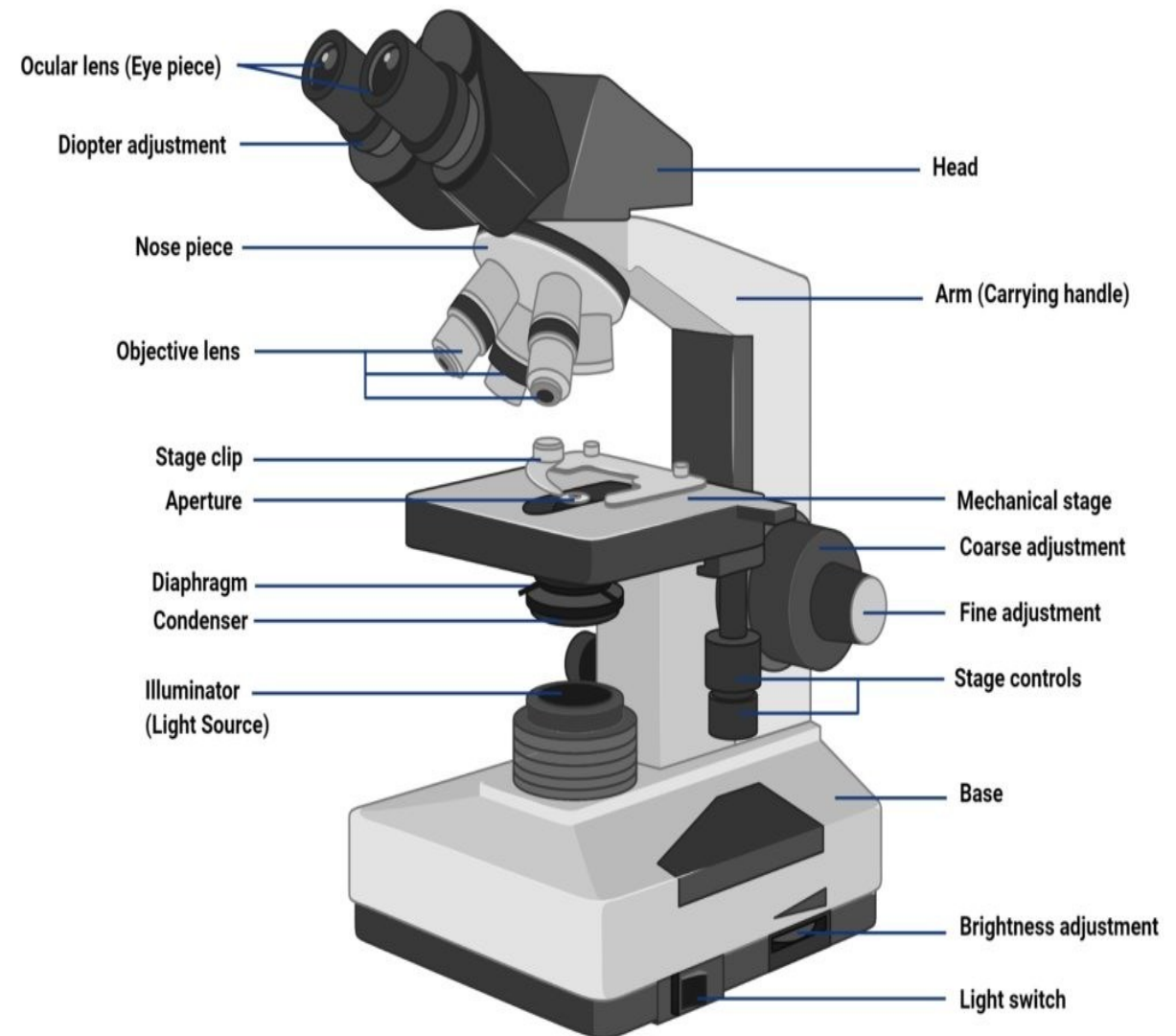
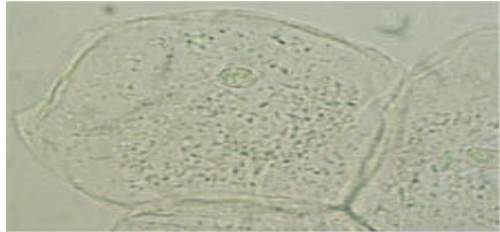
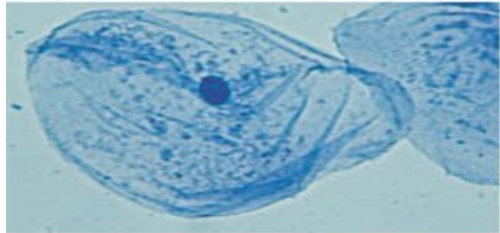
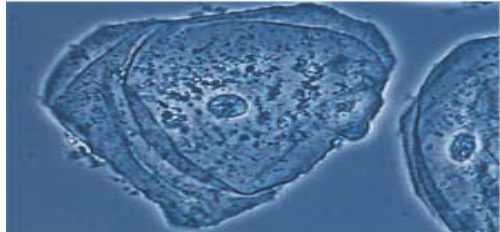


Figure: Parts of a microscope, Image Copyright © Sagar Aryal, www.microbenotes.com

- Three important parameters of microscopy:
 - *Magnification*, the ratio of an object's image size to its real size
 - *Resolution*, the measure of the clarity of the image, or the minimum distance of two distinguishable points
 - *Contrast*, visible differences in brightness between parts of the sample

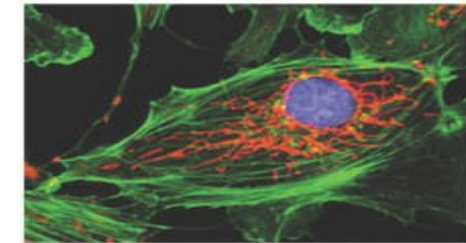
Different Methods For Enhancing Visualization Of Cellular Structures

TECHNIQUE	RESULT
(a) Brightfield (unstained specimen). Passes light directly through specimen. Unless cell is naturally pigmented or artificially stained, image has little contrast. [Parts (a)–(d) show a human cheek epithelial cell.]	 50 μm
(b) Brightfield (stained specimen). Staining with various dyes enhances contrast, but most staining procedures require that cells be fixed (preserved).	
(c) Phase-contrast. Enhances contrast in unstained cells by amplifying variations in density within specimen; especially useful for examining living, unpigmented cells.	

- (d) **Differential-interference-contrast (Nomarski).**
Like phase-contrast microscopy, it uses optical modifications to exaggerate differences in density, making the image appear almost 3D.

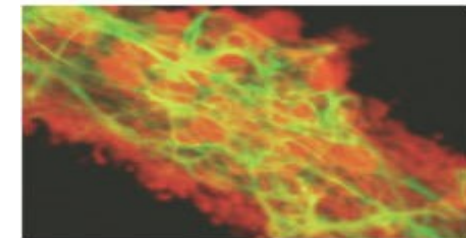
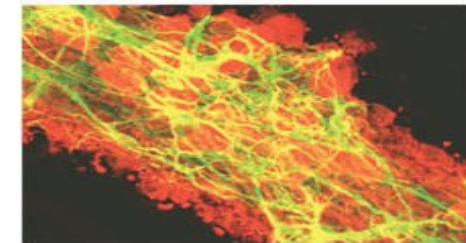


- (e) **Fluorescence.** Shows the locations of specific molecules in the cell by tagging the molecules with fluorescent dyes or antibodies. These fluorescent substances absorb ultraviolet radiation and emit visible light, as shown here in a cell from an artery.



50 μm

- (f) **Confocal.** Uses lasers and special optics for “optical sectioning” of fluorescently-stained specimens. Only a single plane of focus is illuminated; out-of-focus fluorescence above and below the plane is subtracted by a computer. A sharp image results, as seen in stained nervous tissue (top), where nerve cells are green, support cells are red, and regions of overlap are yellow. A standard fluorescence micrograph (bottom) of this relatively thick tissue is blurry.



50 μm

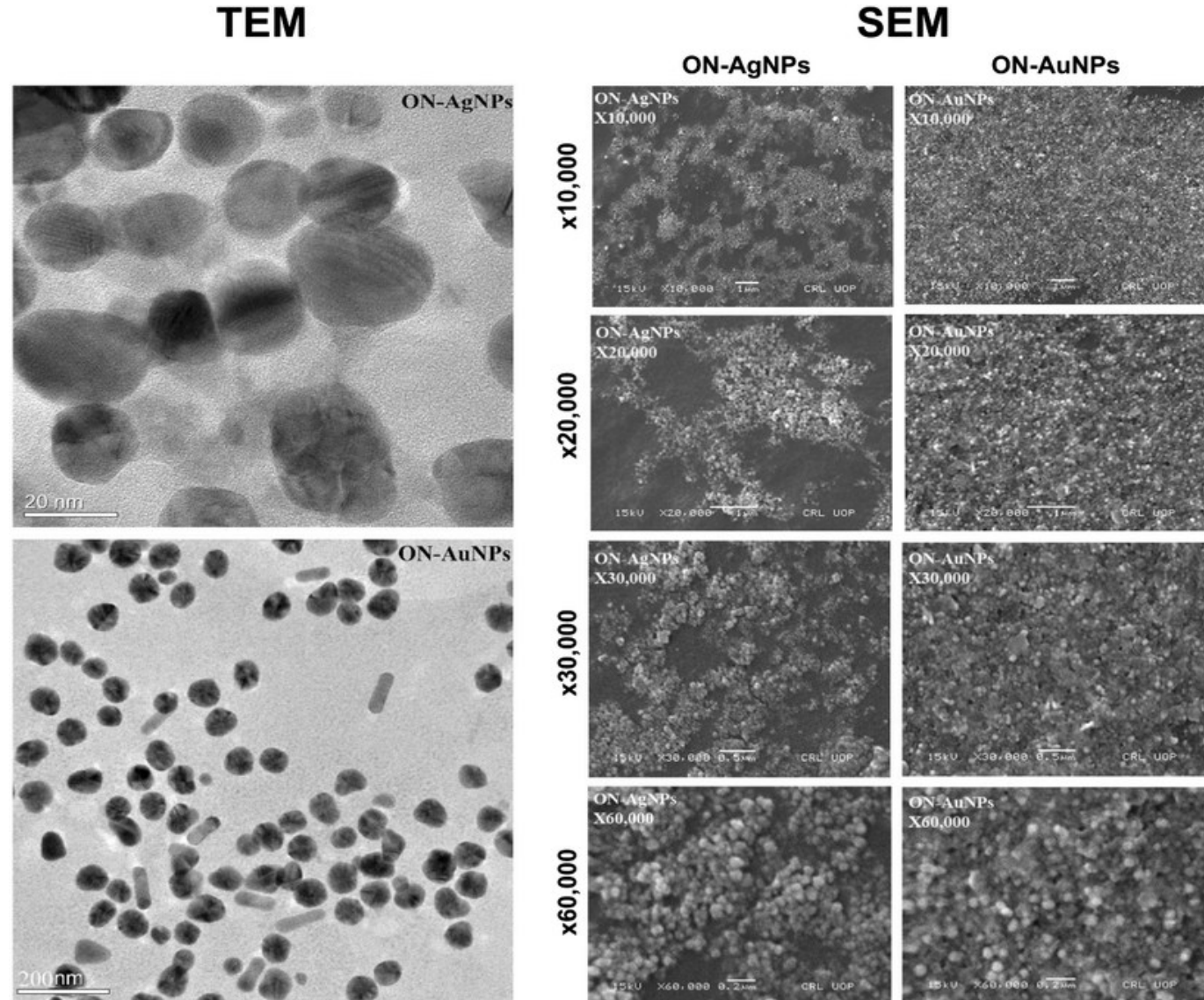
- Electron microscopes (EMs)
 - Focus a beam of electrons through a specimen (TEM) or onto its surface (SEM)
 - 2 types:
 - Transmission electron microscope
 - Scanning electron microscope

Differences between SEM and TEM

TEM	SEM
Electron beam passes through thin sample.	Electron beam scans over surface of sample.
Specially prepared thin samples are supported on TEM grids.	Sample can be any thickness and is mounted on an aluminum stub.
Specimen stage halfway down column.	Specimen stage in the chamber at the bottom of the column.
Image shown on fluorescent screen.	Image shown on TV monitor.
Image is a two dimensional projection of the sample.	Image is of the surface of the sample

Multifunctional theranostic applications of biocompatible green-synthesized colloidal nanoparticles.
Applied microbiology and biotechnology.

Ovais, Muhammad & Khalil, Ali & Raza, Abida & Islam, Nazar Ul & Ayaz, Muhammad & Muthupandian, Saravanan & Ali, Muhammad & Ahmad, Irshad & Shahid, Muhammad & Shinwari, Zabta. (2018).

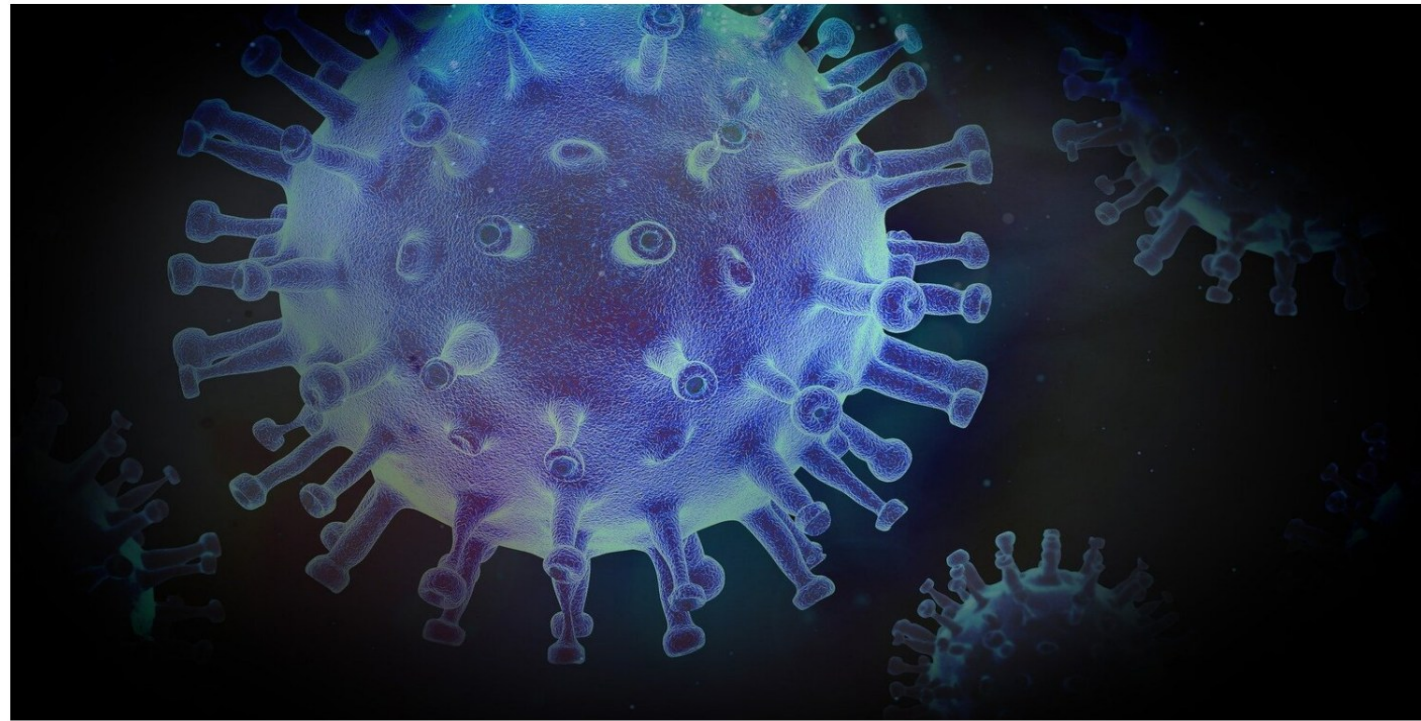


[How cryo-EM aids the search for a COVID-19 vaccine by giving insight into the spike-protein structure — Linkam Scientific](#)

24 April 2020

How cryo-EM aids the search for a COVID-19 vaccine by giving insight into the spike-protein structure

- **Cryo-electron microscopy (cryo-EM)** allows preservation of specimens at very low temperatures
- This allows visualization of structures in their cellular environment, with no need for preservatives
- This method is used to complement X-ray crystallography in revealing protein complexes and subcellular structures
- Microscopes are important tools of cytology, the study of cell structure

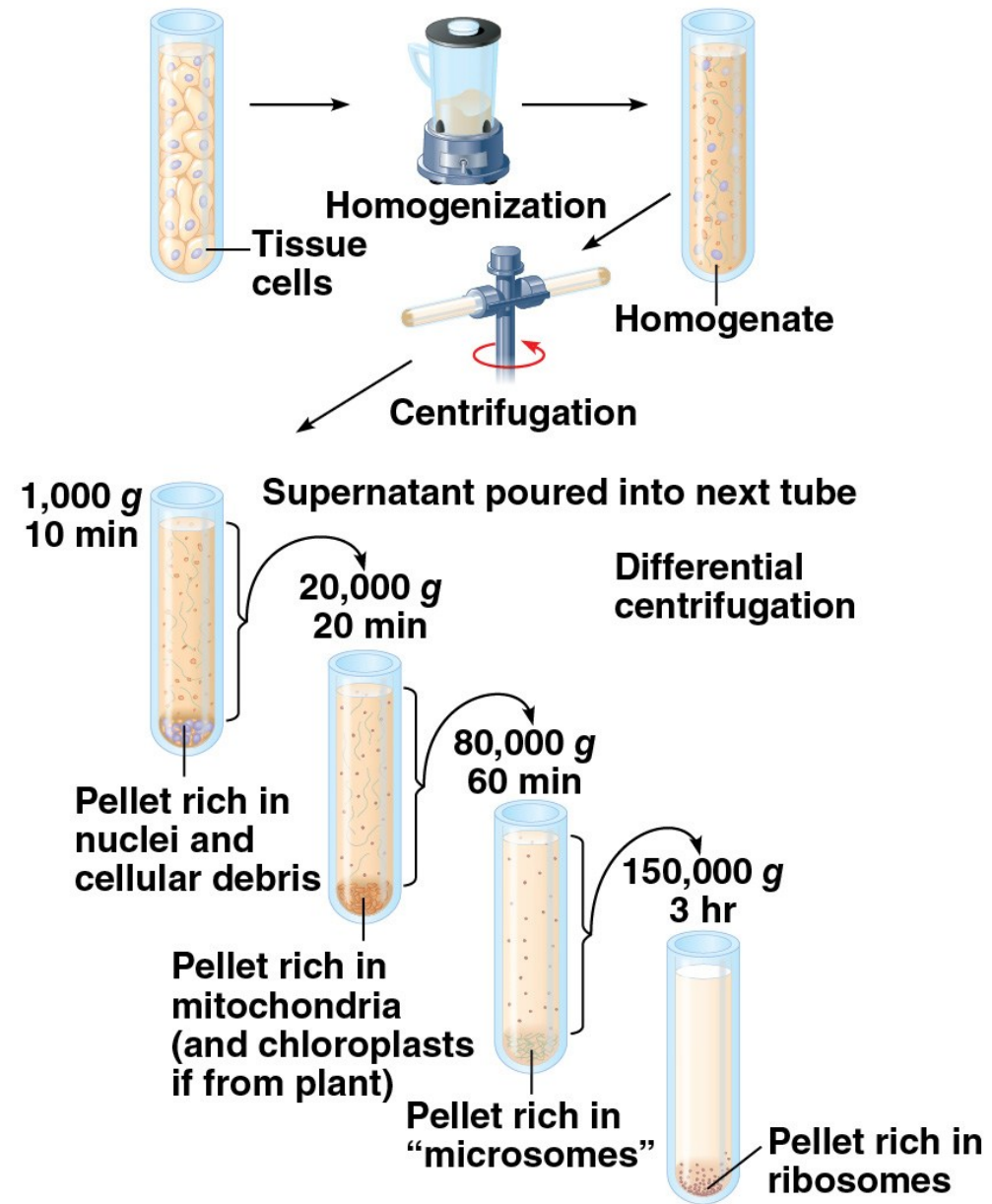


SARS-CoV-2 is the virus responsible for the COVID-19 pandemic. Read how cryo-EM can help understand its properties.

Cell Fractionation

- **Cell fractionation** takes cells apart and separates the major organelles from one another
- Centrifuges fractionate cells into their component parts (differential centrifugation)
- Cell fractionation enables scientists to determine the functions of organelles
- Biochemistry and cytology help correlate cell function with structure

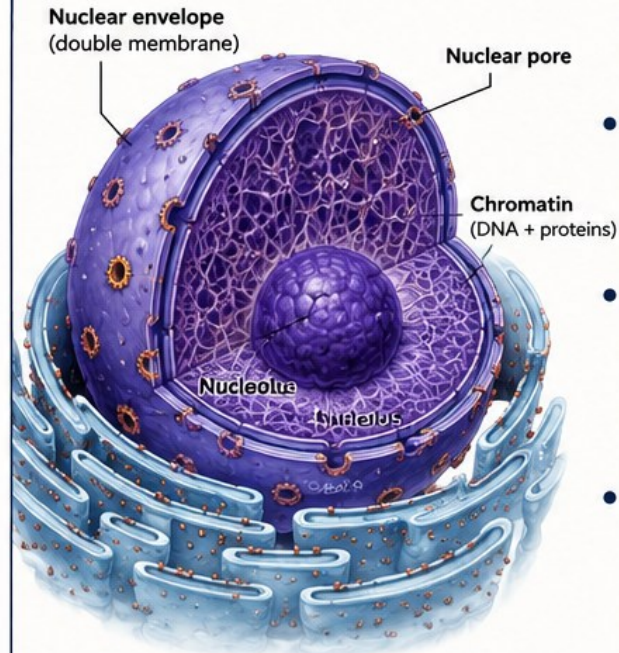
Research method: cell fractionation



Organelle 1 – The Nucleus & Ribosomes: Genetic Control & Synthesis



THE NUCLEUS: Information Central



- **Nuclear Envelope:** Increased in a complex double membrane that is continuous with the endoplasmic reticulum.
- **Nuclear Pores:** Macromolecular pore complexes perforate the envelope, regulating inbound and outbound transport of macromolecules (RNA, proteins).
- **Chromatin Organization:** DNA is associated with basic histone proteins, organizing into fibrous chromatin that condenses into visible chromosomes during division.
- **Nucleolus (Ribosome Construction):** Dense, non-membrane sub-compartment focused on synthesizing ribosomal RNA (rRNA) and assembling large/small ribosomal subunits.



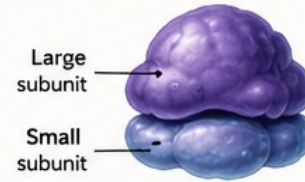
KEY TAKEAWAY

The nucleus stores and protects genetic information (DNA) and controls cell activities through gene expression.

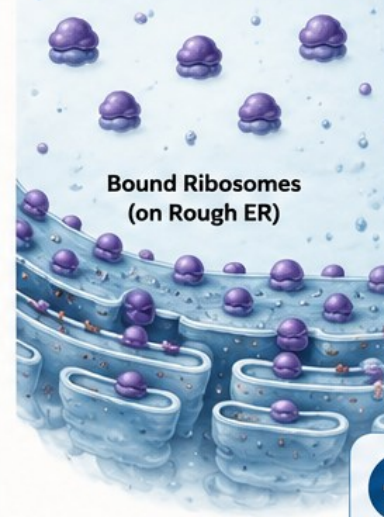


RIBOSOMES: Protein Factories

Ribosome Structure



Free Ribosomes



Bound Ribosomes (on Rough ER)

- **Complex Structure:** Composed of ribosomal RNA (rRNA) transcripts and catalytic polypeptide proteins.
- **Cellular Localization:** Floats freely in cytosol or sits bound to the rough ER outer membrane.
- **Functional Distinction:**
 - **Free Ribosomes:** Chiefly construct intracellular proteins designed for use within the active cytosol (e.g., metabolic enzymes).
 - **Bound Ribosomes:** Synthesize proteins destined for membrane insertion, lysosomal packaging, or extracellular secretion.
- **Evolutionary Preservation:** Ribosomal structures are highly conserved across all branches of life, reflecting universal evolutionary lineage.



KEY TAKEAWAY

Ribosomes are the universal machinery that translates genetic instructions into proteins.

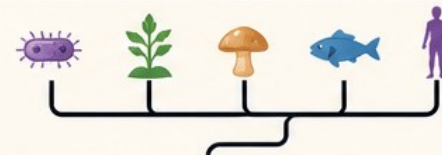


DID YOU KNOW?

Ribosomes are found in all living cells, from bacteria to humans!

Why so important?

The nucleus directs what to build. Ribosomes build the proteins that keep the cell—and you—alive.



Highly conserved across evolution— a universal feature of life.



Organelle 2 – Rough Endoplasmic Reticulum:

The Membrane & Secretory Factory



STRUCTURAL CONTINUITY:

Features an expansive labyrinth of membranous tubules and flattened sacs (cisternae) continuous with the nuclear envelope.



SURFACE ANATOMY:

The outer cytoplasmic face is studded with active bound ribosomes, giving it a rough, coarse texture.



POLYPEPTIDE PROCESSING:

Newly synthesized polypeptide chains are co-translationally translocated directly into the internal ER lumen.



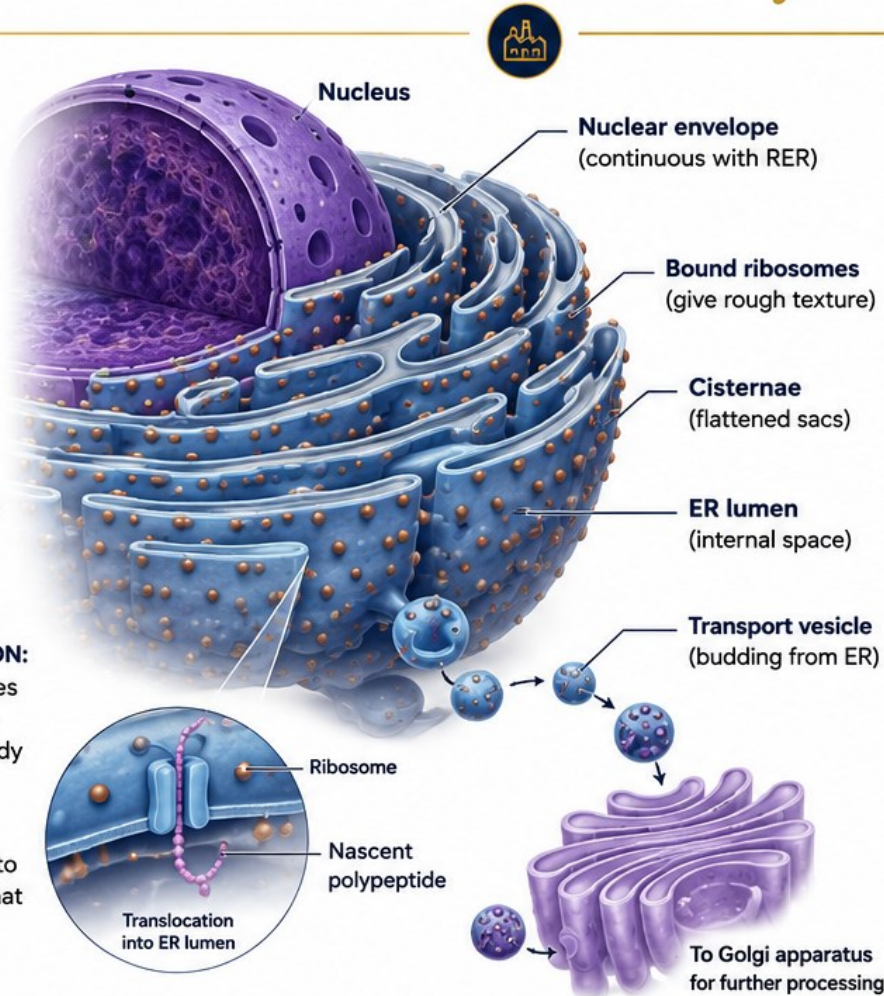
GLYCOPROTEIN CONSTRUCTION:

Within the lumen, carbohydrates are enzymatically appended to proteins, forming transport-ready glycoproteins.



SECRETORY PACKAGING:

ER packages secretory cargo into membrane transport vesicles that pinch off from transitional ER regions.



BIO-FACTORY ROLE



MEMBRANE FACTORY OF THE CELL:

The Rough ER continuously expands its own membrane by synthesizing phospholipids and membrane-bound proteins.

As transport vesicles pinch off to target other organelles, the ER self-replenishes its surface area.



HIGH SECRETORY DENSITIES:

Cells highly specialized in protein secretion (e.g., human pancreatic cells secreting insulin) are packed with massive networks of Rough ER.



Rough ER



Transport vesicle



Secretion to other locations



DID YOU KNOW?

Rough ER is especially abundant in cells that make and secrete large amounts of proteins, such as plasma cells and pancreatic cells!



KEY TAKEAWAY

Rough ER is the cell's manufacturing and processing unit—building proteins, modifying them, and packaging them for delivery where they are needed.

WHERE IS IT FOUND?

In all eukaryotic cells, more extensive in cells that synthesize or secrete large quantities of proteins (e.g., secretory gland cells).



The endomembrane system regulates protein traffic and performs metabolic functions

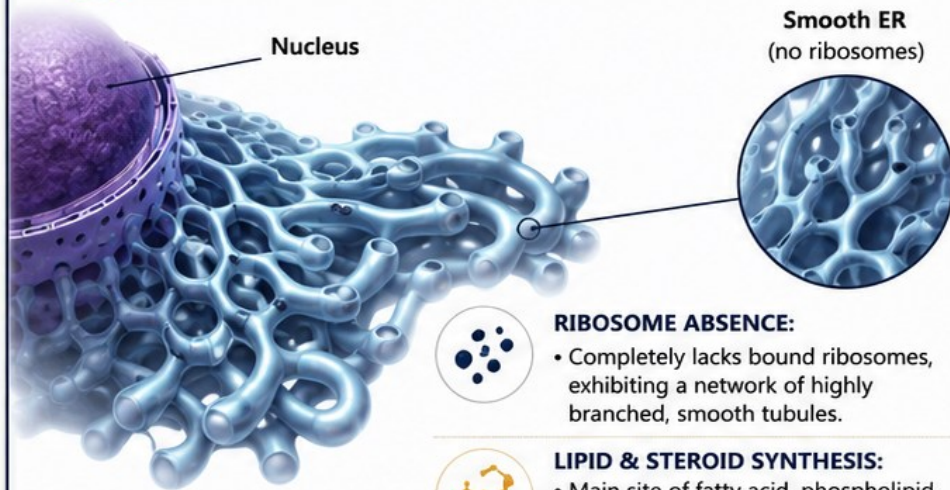
- The **endomembrane system** consists of
 - Nuclear envelope
 - Endoplasmic reticulum
 - Golgi apparatus
 - Lysosomes
 - Vacuoles
 - Plasma membrane
- These components are either continuous or connected via transfer by **vesicles**

Organelle 3 – Smooth Endoplasmic Reticulum:

Lipid Synthesis & Detoxification



SMOOTH ER: Tubular Metabolic Hub



RIBOSOME ABSENCE:

- Completely lacks bound ribosomes, exhibiting a network of highly branched, smooth tubules.



LIPID & STEROID SYNTHESIS:

- Main site of fatty acid, phospholipid, and steroid hormone (such as estrogen, progesterone, and testosterone) synthesis.



CARBOHYDRATE METABOLISM:

- Enzymes in the smooth ER membrane catalyze glycogenolysis (especially in liver cells), converting stored glycogen to free glucose.



DYNAMIC MEMBRANE EXPANSION:

- Upregulates membranes production rapidly under high metabolic or chemical demands.



DETOXIFICATION & ION STORAGE



CHEMICAL DETOXIFICATION IN LIVER CELLS:

Enzymes (like Cytochrome P450) enzymatically detoxify barbiturates, alcohol, and toxins by appending hydroxyl (-OH) groups. This vastly increases the substance's solubility in water, facilitating rapid excretion from the kidneys.



CALCIUM ION (Ca²⁺) SEQUESTRATION:

The smooth ER acts as a specialized calcium pump, actively sequestering calcium ions from the cytosol.



ROLE IN MUSCLE CELLS:

In muscle cells (sarcoplasmic reticulum), the release of this Ca²⁺ is the molecular trigger for muscle contraction.

KEY FUNCTION FLOW



Smooth ER (SER)



Detoxify toxins or store Ca²⁺



Increase solubility in water



Excretion via kidneys / Trigger muscle contraction



DID YOU KNOW?

Smooth ER is abundant in liver cells and steroid-producing cells (e.g., in the ovaries and testes) because of their intense roles in detoxification and hormone synthesis!



KEY TAKEAWAY

Smooth ER is the cell's metabolic and detox hub—synthesizing lipids and steroids, detoxifying harmful substances, and managing calcium levels for vital cellular functions.

The Endoplasmic Reticulum: Biosynthetic Factory

- The **endoplasmic reticulum (ER)** accounts for more than half of the total membrane in many eukaryotic cells
- The ER membrane is continuous with the nuclear envelope
- There are two distinct regions of ER:
 - **Smooth ER**, which lacks ribosomes
 - **Rough ER**, whose surface is studded with ribosomes

Functions of Rough ER

- The rough ER
 - Has bound ribosomes, which secrete **glycoproteins** (proteins covalently bonded to carbohydrates)
 - Distributes **transport vesicles**, secretory proteins surrounded by membranes
 - Is a membrane factory for the cell

Functions of Smooth ER

- The smooth ER
 - Synthesizes lipids
 - Detoxifies drugs and poisons
 - Stores calcium ions

Organelle 4 – The Golgi Apparatus: Shipping & Receiving Center



THE GOLGI APPARATUS: Structure & Functions



CISTERNAE ARCHITECTURE:

Composed of a series of distinct, flattened membranous sacs called cisternae, arranged like a stack of pita bread.



STRUCTURAL POLARITY:

Features two highly polarized faces representing a directional conveyor belt:

- **Cis Face (Receiving):** Positioned near the ER; fuses with arriving transport vesicles to receive cargo.
- **Trans Face (Shipping):** Faces the plasma membrane; buds off secretory vesicles to deliver finalized products.



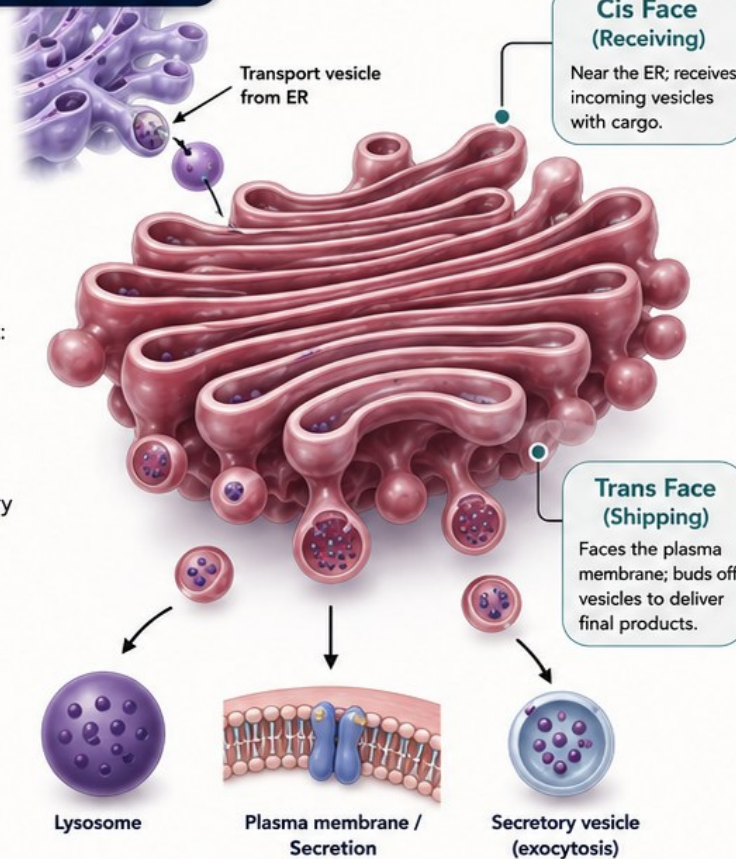
CHEMICAL MODIFICATION:

Modifies ER glycoproteins by trimming and replacing sugar monomers, creating complex oligosaccharides.



POLYSACCHARIDE SYNTHESIS:

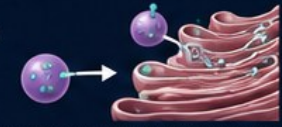
Manufactures non-cellulose structural pectins for the plant cell wall.



THE CONVEYOR BELT FLOW

1

Vesicles bud from transition ER and arrive at the Cis-Golgi face.



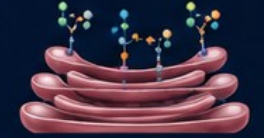
2

Vesicles fuse, dumping cargo proteins into the first cisternae.



3

Proteins migrate through medial cisternae, undergoing glycosylation adjustments.



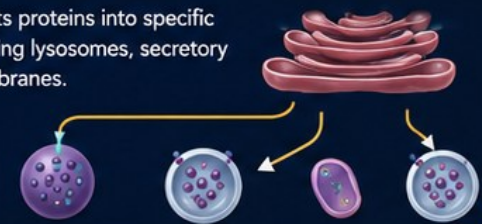
4

Molecular tags (like phosphate tags on lysosomal enzymes) are appended to direct routing.



5

Trans-face sorts proteins into specific vesicles targeting lysosomes, secretory paths, or membranes.



DID YOU KNOW?

The Golgi apparatus was discovered by Camillo Golgi in 1898 while studying nerve cells under the microscope!



KEY TAKEAWAY

The Golgi apparatus acts as the cell's logistics center—modifying, sorting, and packaging proteins and lipids to ensure they reach the right destination at the right time!



Receives cargo



Modifies & processes



Tags & sorts



Ships to the right place

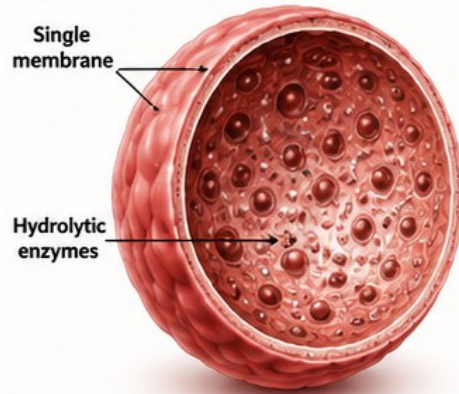
The Golgi Apparatus: Shipping and Receiving Center

- The **Golgi apparatus** consists of flattened membranous sacs called cisternae
- The Golgi apparatus
 - Modifies products of the ER
 - Manufactures certain macromolecules
 - Sorts and packages materials into transport vesicles

Organelle 5 – Lysosomes & Vacuoles: Digestion & Cellular Storage

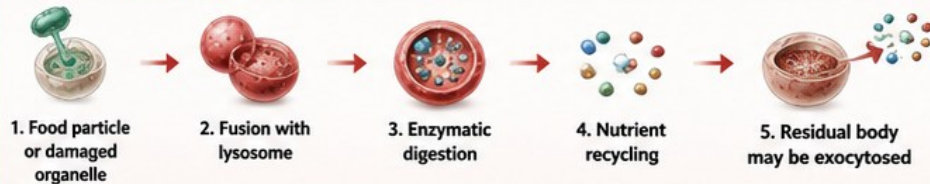


LYSOSOMES: Intracellular Digestion



- **Acidic Membranous Sacs:**
Enclose active hydrolytic enzymes designed to digest macromolecules.
- **Optimized Acidic pH:**
Internal pH is highly acidic (~4.5–5.0), maintained by active H^+ proton pumps.
- **Double Safeguard Mechanism:**
If a lysosome ruptures, the enzymes are inactive in the neutral pH (~7.2) of the cytosol. However, massive rupture causes cellular autolysis.
- **Phagocytosis:**
Fuses with food vacuoles to digest engulfed nutrients or invading pathogens.
- **Autophagy:**
Encloses damaged organelles, recycling their core molecules to continuously renew cellular health.

HOW LYSOSOMES WORK

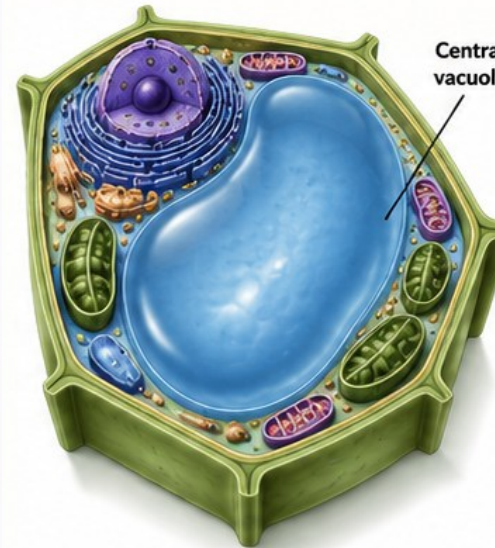


DID YOU KNOW?

Lysosomes were discovered by Christian de Duve in 1955, who received the Nobel Prize in Physiology or Medicine in 1974 for his discoveries on lysosomes and peroxisomes!



VACUOLES: Dynamic Storage Compartments



- **Membrane Boundary:**
Bound by a single selective membrane known as the tonoplast.
- **Food Vacuoles:**
Formed during endocytic phagocytosis to store internalized nutrients.
- **Contractile Vacuoles:**
Found in freshwater protists (e.g., Paramecium); act as bilge pumps to pump out excess water for active osmoregulation.
- **Plant Central Vacuole:**
Holds reserves of organic molecules, water, toxic defense compounds, and pigments.
- **Turgor Control:**
Directly maintains plant structural stiffness by pressing the cytoplasm against the rigid cell wall.

TYPES OF VACUOLES



Contractile vacuole (Protists)
Removes excess water



Food vacuole (Animal cells)
Stores and digests engulfed nutrients



Central vacuole (Plant cells)
Stores water, nutrients, pigments, and wastes



Storage vacuole (Fungi)
Stores nutrients and ions



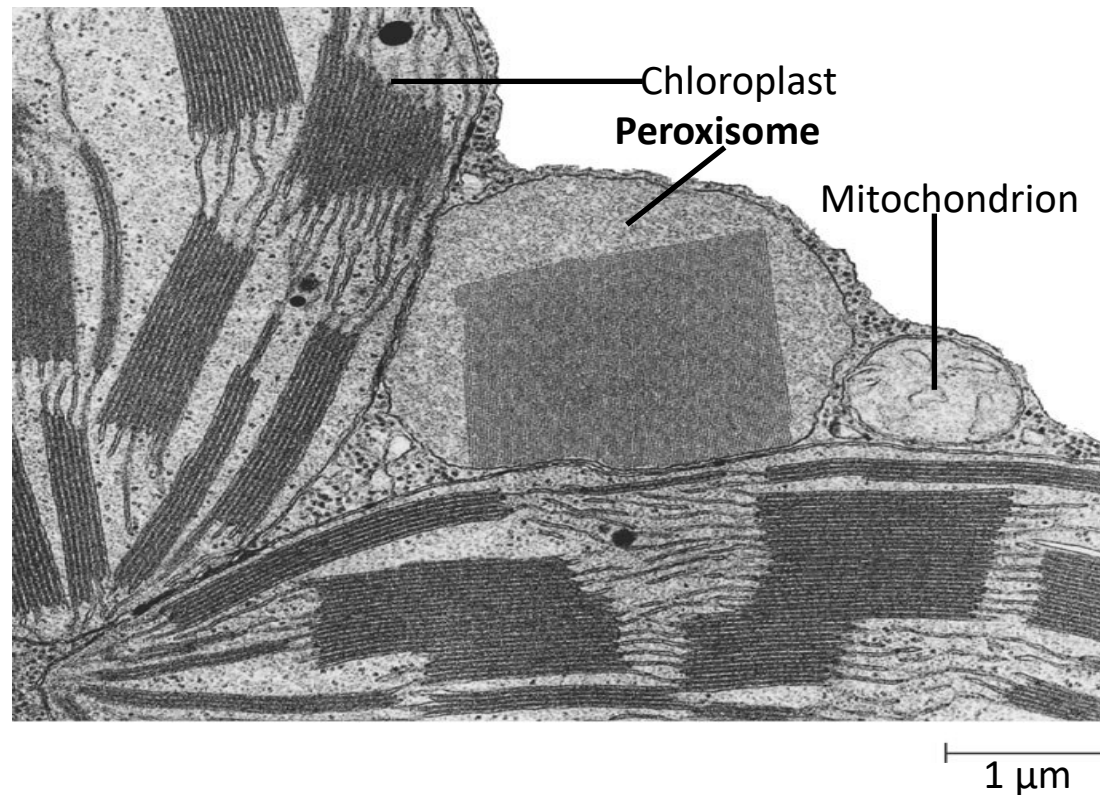
KEY TAKEAWAY

Lysosomes are the cell's digestion center, breaking down materials and recycling building blocks. Vacuoles are versatile storage compartments that maintain balance, store resources, and support cell structure and survival.



Peroxisomes

- Present of catalase
- Converts hydrogen peroxide (H_2O_2) to water & O_2
- Detoxify alcohol in liver cells



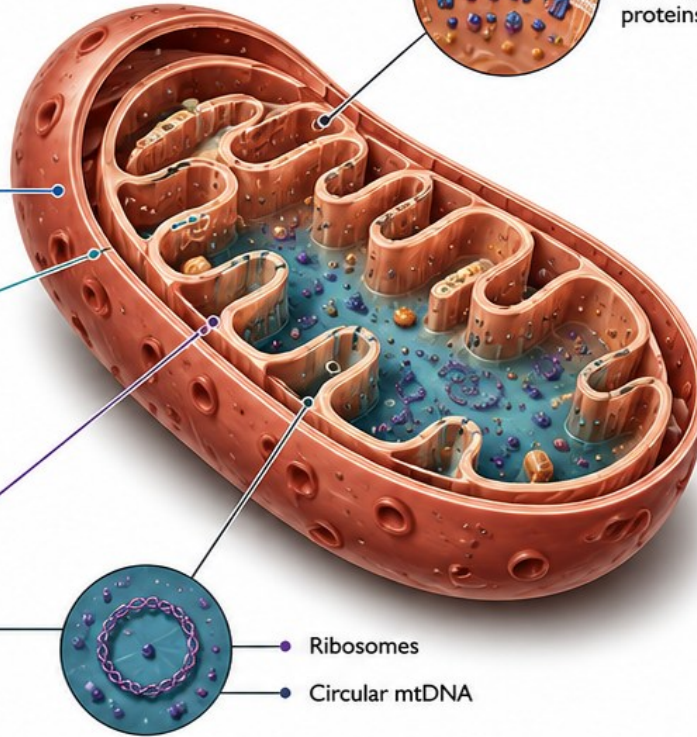
Organelle 6 – Mitochondria:

Chemical Powerhouse and Cellular Respiration



THE MITOCHONDRIA: Structure & Function

- **Chemical Respiration Site:**
Generates metabolic energy (ATP) from organic molecules (glucose, fatty acids) in the presence of oxygen.
- **Double-Membrane Envelope:**
 - **Smooth Outer Membrane:**
Permeable to small molecules via porin channels.
 - **Folded Inner Membrane (Cristae):**
Convolved folds increase surface area, packing millions of Electron Transport Chain (ETC) proteins.
- **Two Fluid Compartments:**
 - **Intermembrane Space:**
High proton (H^+) concentration generated by electron transport.
 - **Mitochondrial Matrix:**
Inner fluid containing Krebs cycle enzymes, ribosomes, and circular mitochondrial DNA (mtDNA).



Electron Transport Chain (ETC) proteins

Ribosomes
Circular mtDNA



METABOLIC ADAPTATION



High Surface Area Advantage:

By folding the inner membrane into Cristae, the mitochondrion maximizes membrane space for ATP Synthase complexes.



Autonomous Biology:

Contains its own circular DNA and 70S ribosomes, allowing it to synthesize a portion of its own respiratory proteins.



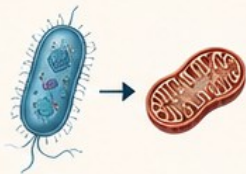
Dynamic Populations:

Cells with high energy demands (e.g., cardiac muscle cells, active liver hepatocytes) contain thousands of mitochondria.



DID YOU KNOW?

Mitochondria are believed to have evolved from ancient aerobic bacteria that were engulfed by early cells over 1.5 billion years ago—a true example of endosymbiosis!



KEY TAKEAWAY

Mitochondria are the cell's power stations—converting nutrients and oxygen into ATP. Their unique structure, dual membranes, and own DNA make them highly efficient and partially self-sufficient organelles adapted for energy production.



Organelle 7 – Chloroplasts & Plastids:

Solar Converters of Autotrophs



PHOTOSYNTHETIC ENERGY TRANSDUCERS:

Capture solar radiation and convert it to chemical energy (sugars) using water and carbon dioxide.



CHLOROPHYLL CHEMISTRY:

Contain green chlorophyll pigments embedded in internal thylakoid membranes to absorb light.



THREE MEMBRANE COMPARTMENTS:

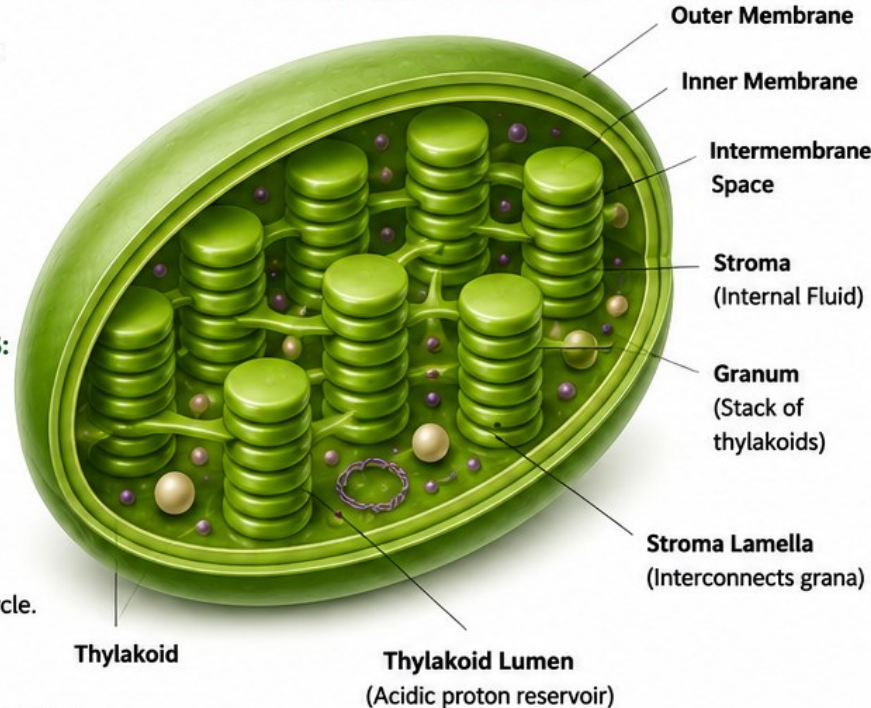
- **Intermembrane Space:** Segregates the double envelope.
- **Thylakoid Lumen:** Acidic proton reservoir during light reactions.
- **Stroma (Internal Fluid):** Contains circular DNA, 70S ribosomes, starch granules, and enzymes for the Calvin cycle.



MEMBRANOUS STACK SYSTEMS:

- **Thylakoids:** Flattened, interconnected sacs.
- **Granum:** Organized columns of 10 to 100 thylakoids.

CHLOROPLAST STRUCTURE



THE PLASTID FAMILY

Chloroplasts are specialized members of a plant organelle family known as **Plastids**:



AMYLOPLASTS / LEUCOPLASTS:

Colorless plastids found in roots and tubers that synthesize and store starch molecules.



CHROMOPLASTS:

Enriched in carotenoid pigments, creating bright red, orange, and yellow hues in autumn leaves, flowers, and ripe fruits.



DID YOU KNOW?

Chloroplasts are believed to have evolved from ancient cyanobacteria through endosymbiosis over 1.5 billion years ago!



KEY TAKEAWAY

Chloroplasts and other plastids capture light energy or store essential molecules, making them the solar converters and storage specialists that sustain plant life.

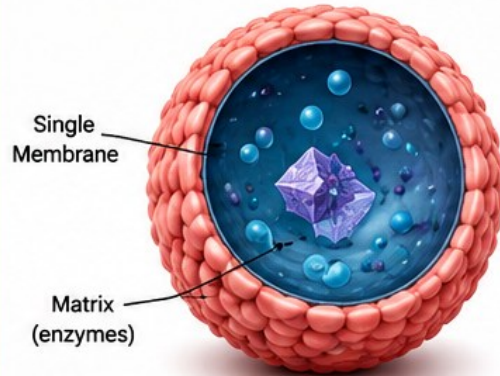


Organelle 8 – Peroxisomes & Cytoskeleton:

Metabolic Safety & Framework



PEROXISOMES: Safety Oxidizers



- **Single-Membrane Metabolizers:**
Bound by a single membrane; transfer hydrogen from compounds to oxygen (O_2), producing toxic hydrogen peroxide (H_2O_2) as a byproduct.
- **Catalase Defense:**
Contain high concentrations of catalase enzymes that instantly convert dangerous H_2O_2 to safe water (H_2O) and oxygen (O_2).
- **Fatty Acid Breakdown:**
Oxidize long-chain fatty acids into smaller molecules, exporting them to mitochondria for respiration.
- **Alcohol Detoxification:**
Enriched in liver cells to detoxify ethanol and other poisons.

CATALASE REACTION (DETOXIFICATION)



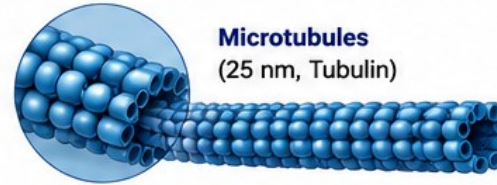
KEY TAKEAWAY

Peroxisomes protect the cell by neutralizing toxins and breaking down fatty acids, while detoxifying harmful compounds and supporting metabolic balance.

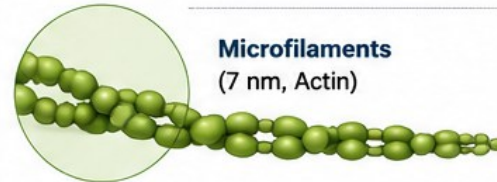


CYTOSKELETON: Structural Highway

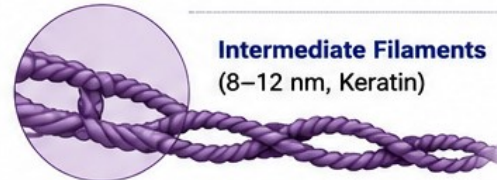
A dynamic network of protein fibers extending throughout the cytoplasm to coordinate shape and transport:



- **Microtubules (25 nm, Tubulin):**
Resist cellular compression; form intracellular transport highways; guide chromosomes during division; construct cilia and flagella.



- **Microfilaments (7 nm, Actin):**
Bear cellular tension; enable cell crawling, amoeboid movement, cytoplasmic streaming, and cleavage furrows.



- **Intermediate Filaments (8–12 nm, Keratin):**
Anchor the nucleus; build the nuclear lamina; highly stable (only in animal cells).



KEY TAKEAWAY

The cytoskeleton provides the structural framework and dynamic tracks that maintain cell shape, support movement, and organize internal transport.

TOGETHER, THEY MAINTAIN CELL HEALTH AND FUNCTION



Detoxify harmful compounds



Support lipid metabolism



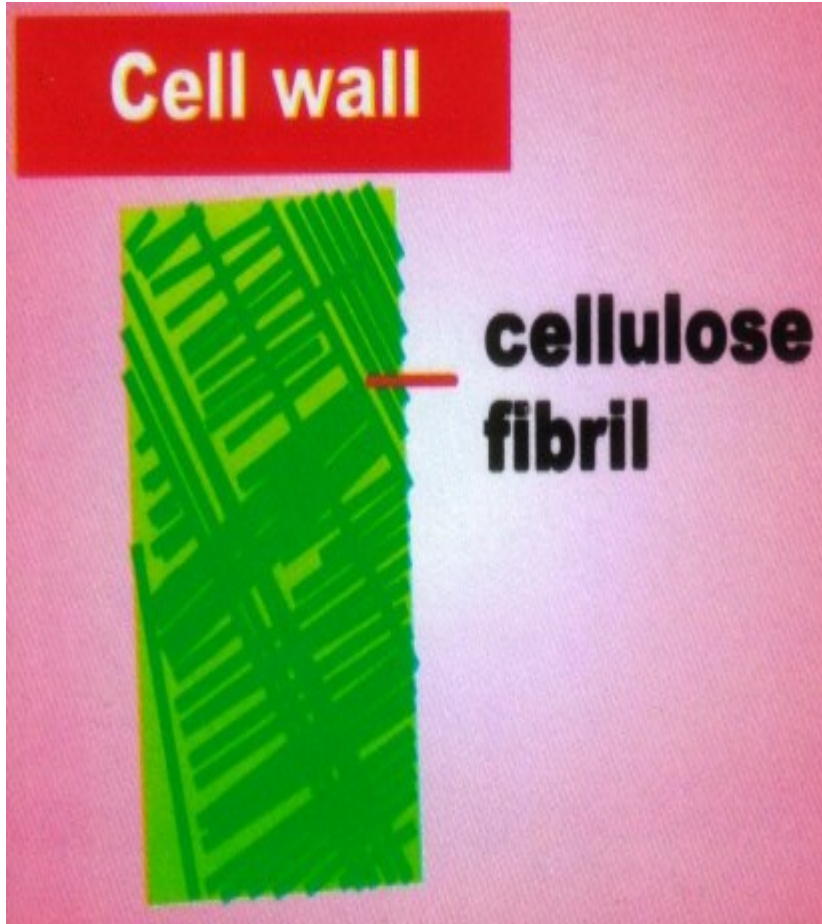
Maintain cellular balance



Enable movement and transport



Organize cell structure



- Maintains the shape of plant cells.
- Provides mechanical strength and support to plant cells.
- Protects plants cells from rupturing due to the movement of excess water into the cells.

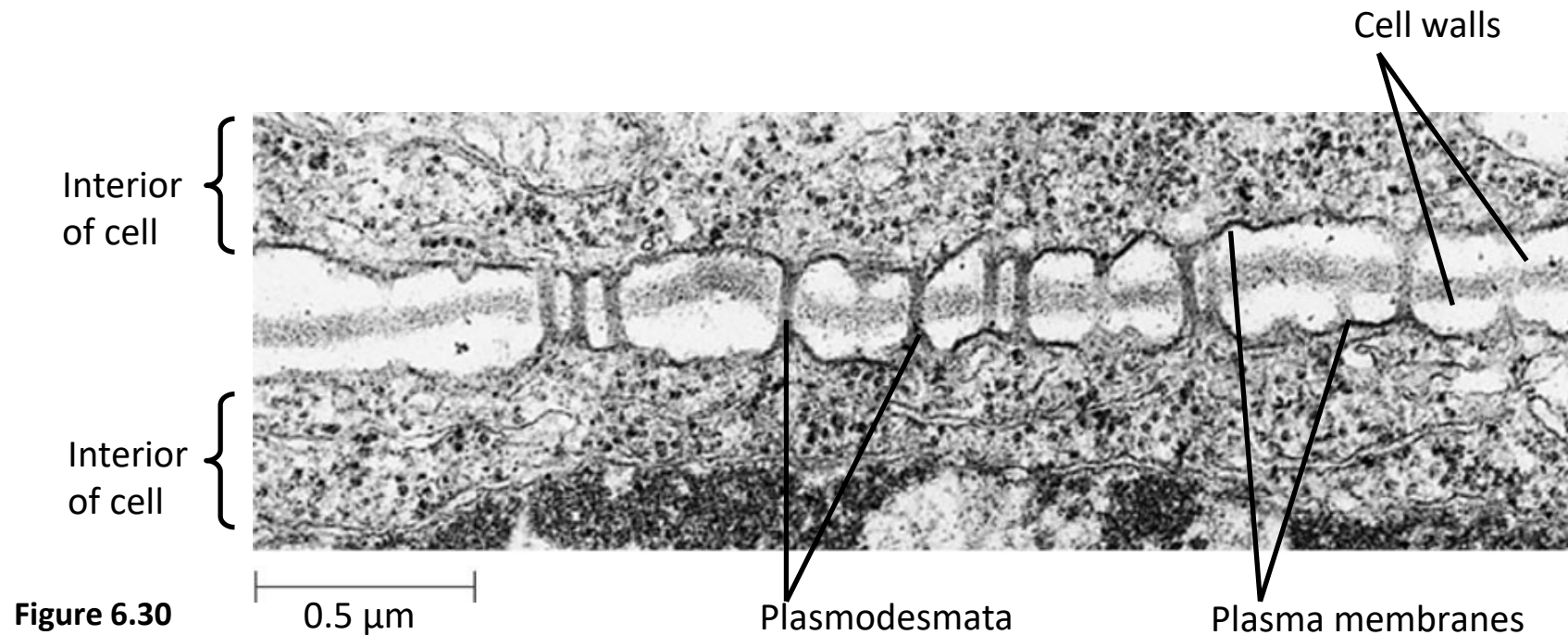
- Plant & algae have cell walls made of cellulose
- Fungi's cell walls are from chitin
- 2 types:
 - Primary cell wall
 - Secondary cell wall

Primary Cell Wall

- Found in young cells & cells that are not highly differentiated such as meristem, parenchyma, collenchyma
- Thin layers, found just outside of the plasma membrane
- Randomly arranged microfibrils of cellulose
- Usually perforated with plasmodesmata for transport

Plants: Plasmodesmata

- Plasmodesmata
 - Are channels that perforate plant cell walls



Secondary Cell Wall

- Harder & usually thicker layer of cell wall
- Form between plasma membrane & primary cell wall
- Regularly arranged microfibrils or bigger microfibrils in parallel rows, perpendicular to those upper or lower layers
- Impregnated with lignin forming a hard & impervious layer

Checkpoints

- How do you connect the cell theory to the evolution of life?
- How does the structure of the nucleus affect how information stored in DNA is used by the cells?

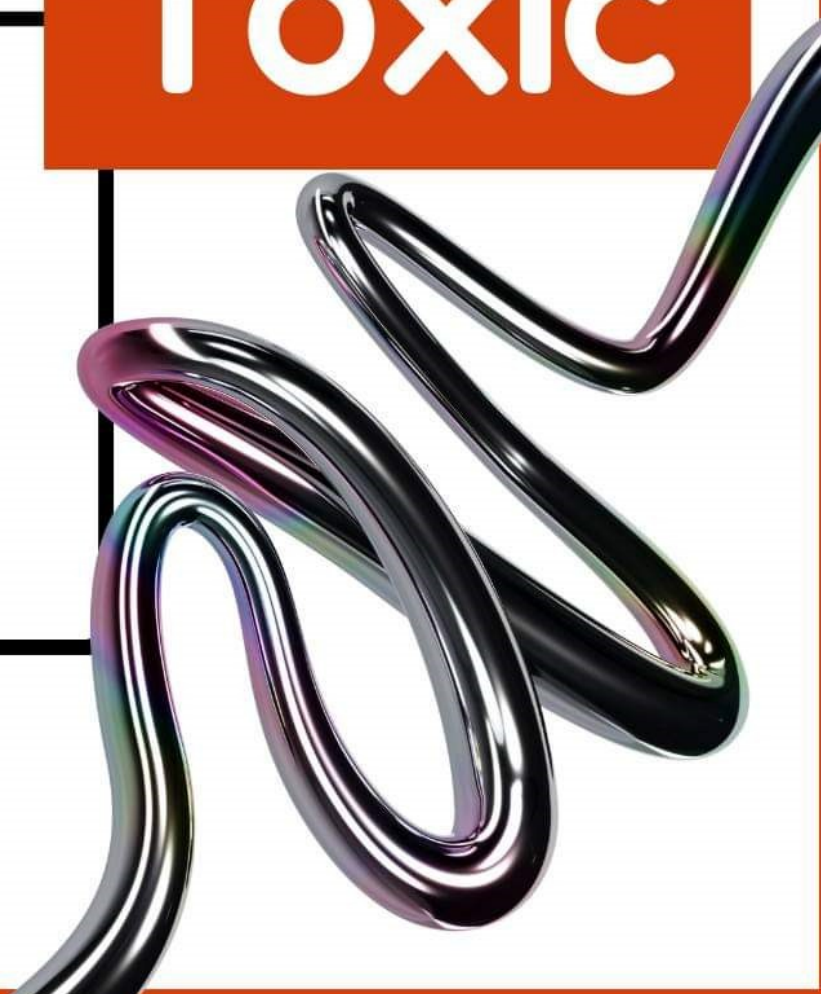


Food for thought!

A **Hug** WITHOUT **U** IS

TOXIC

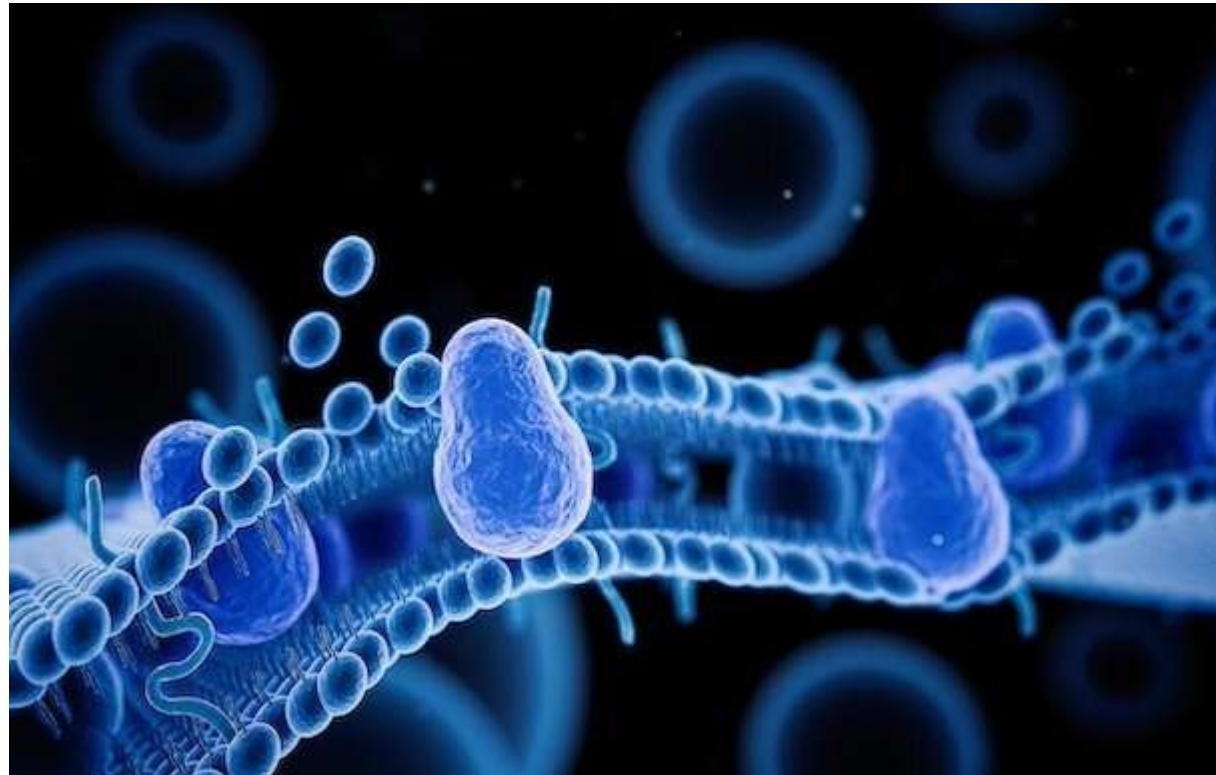
80
Hg
Mercury
200.59



The graphic features a central rectangular box representing a mercury atom. Inside the box, the atomic number '80' is at the top left, the chemical symbol 'Hg' is in the center, and the name 'Mercury' and atomic weight '200.59' are at the bottom. To the right of the box is a red rectangle with the word 'TOXIC' in white. Above the box, the text 'A Hug WITHOUT U IS' is displayed. A wavy, metallic-looking line, resembling a mercury droplet or a stylized electron cloud, curves around the bottom right of the atom box.

THE CELL: Part 2

Siew Ee Ling, PhD



The Thin Line Between Inside and Outside: A Membrane's Tale

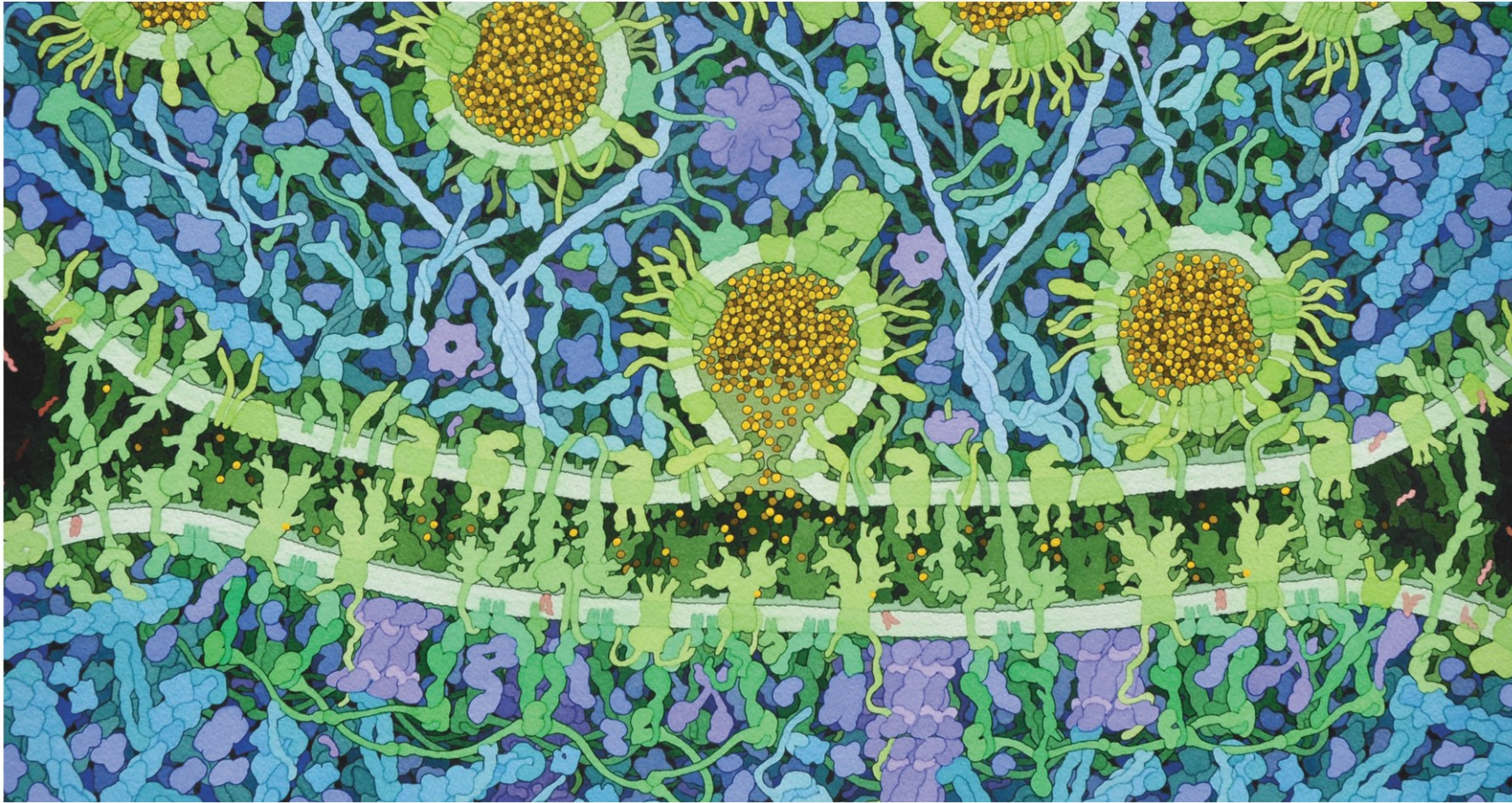
What's the plasma membrane's favorite movie?

A: "The Matrix" - it's all about controlling what gets in and out!



If the plasma membrane had a dating profile, what would be its best quality?

A: "I'm flexible, but I know how to maintain boundaries."

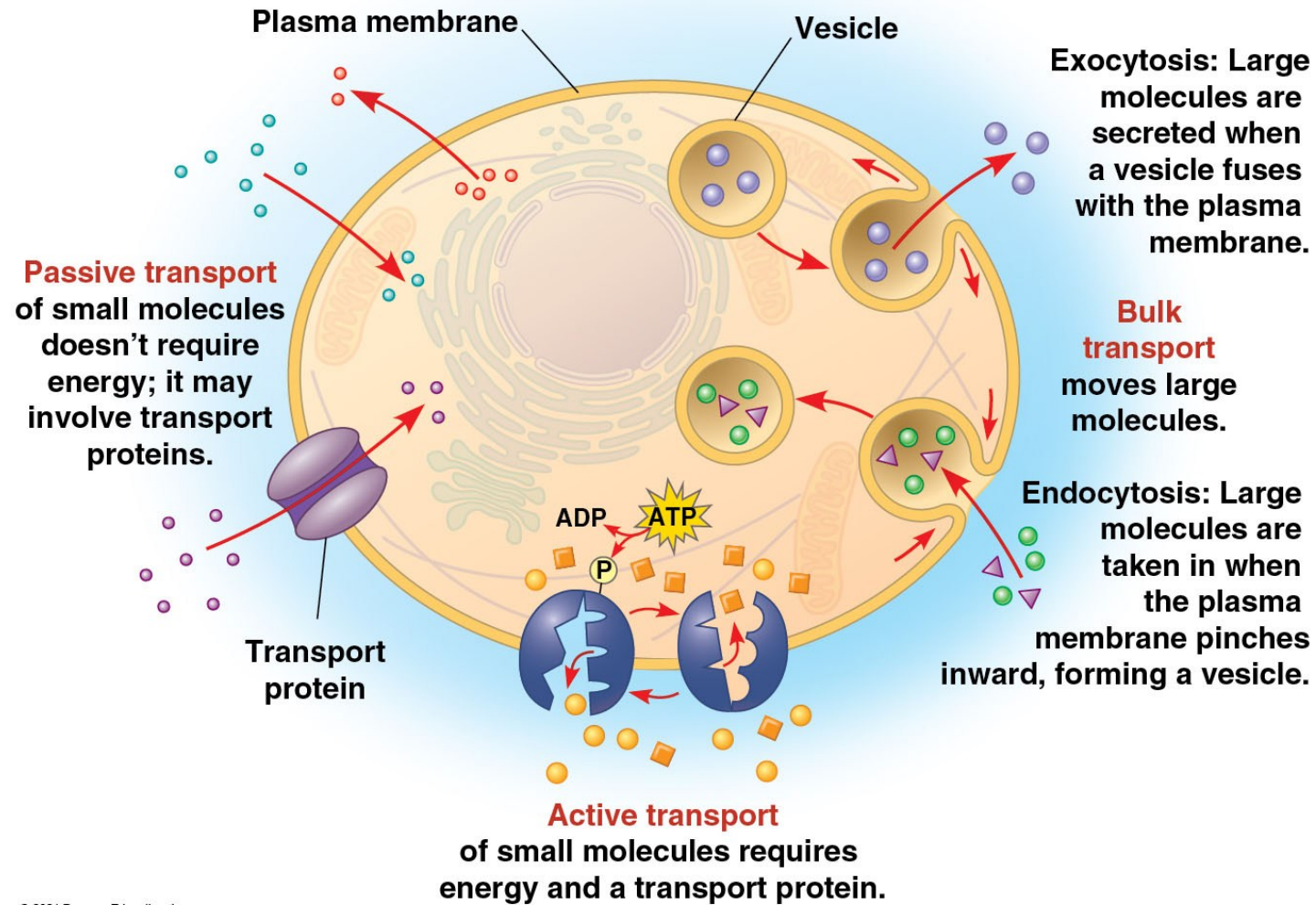


Successful learning relies on communication between brain cells. Here, the vesicles fusing with the plasma membrane of the top cell release molecules (yellow) that bind to membrane proteins (light green) on the surface of the bottom cell, triggering the proteins to change shape. The plasma membrane that surrounds each cell regulates its exchanges with its environment and surrounding cells.

How does the plasma membrane regulate inbound and outbound traffic?

- Some small molecules move across the cell membrane using passive transport—no input of energy—and may require transport proteins
- Some small molecules use active transport, which requires both energy and a transport protein
- Large molecules move in and out, using bulk transport; exocytosis or endocytosis

How does the plasma membrane regulate inbound and outbound traffic?

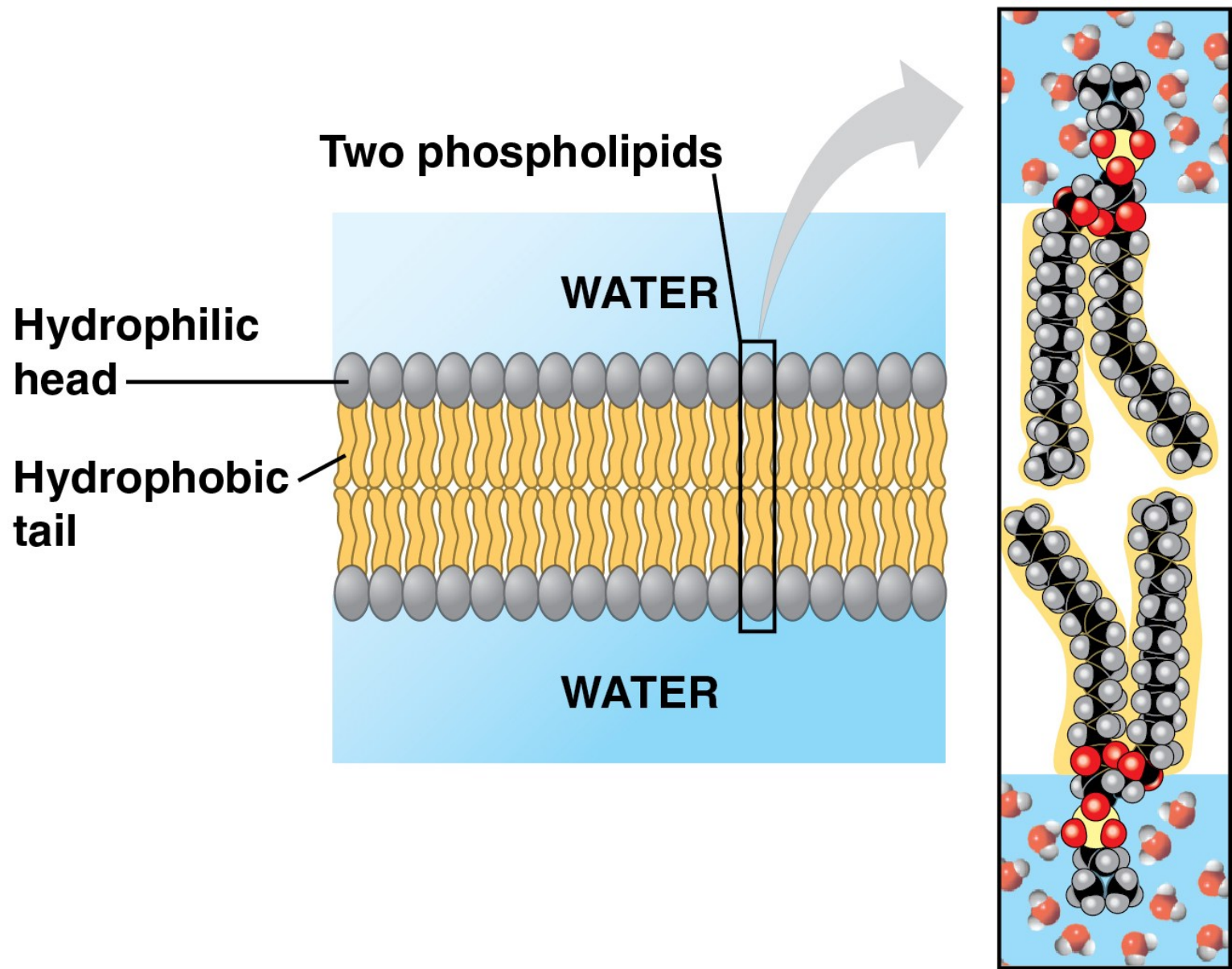


Cellular membranes are fluid mosaics of lipids and proteins

- Lipids and proteins are the main components of membranes, but carbohydrates are also important
- Membranes are composed mainly of phospholipids
- Phospholipids are **amphipathic** molecules, containing hydrophobic (“water-fearing”) and hydrophilic (“water-loving”) regions

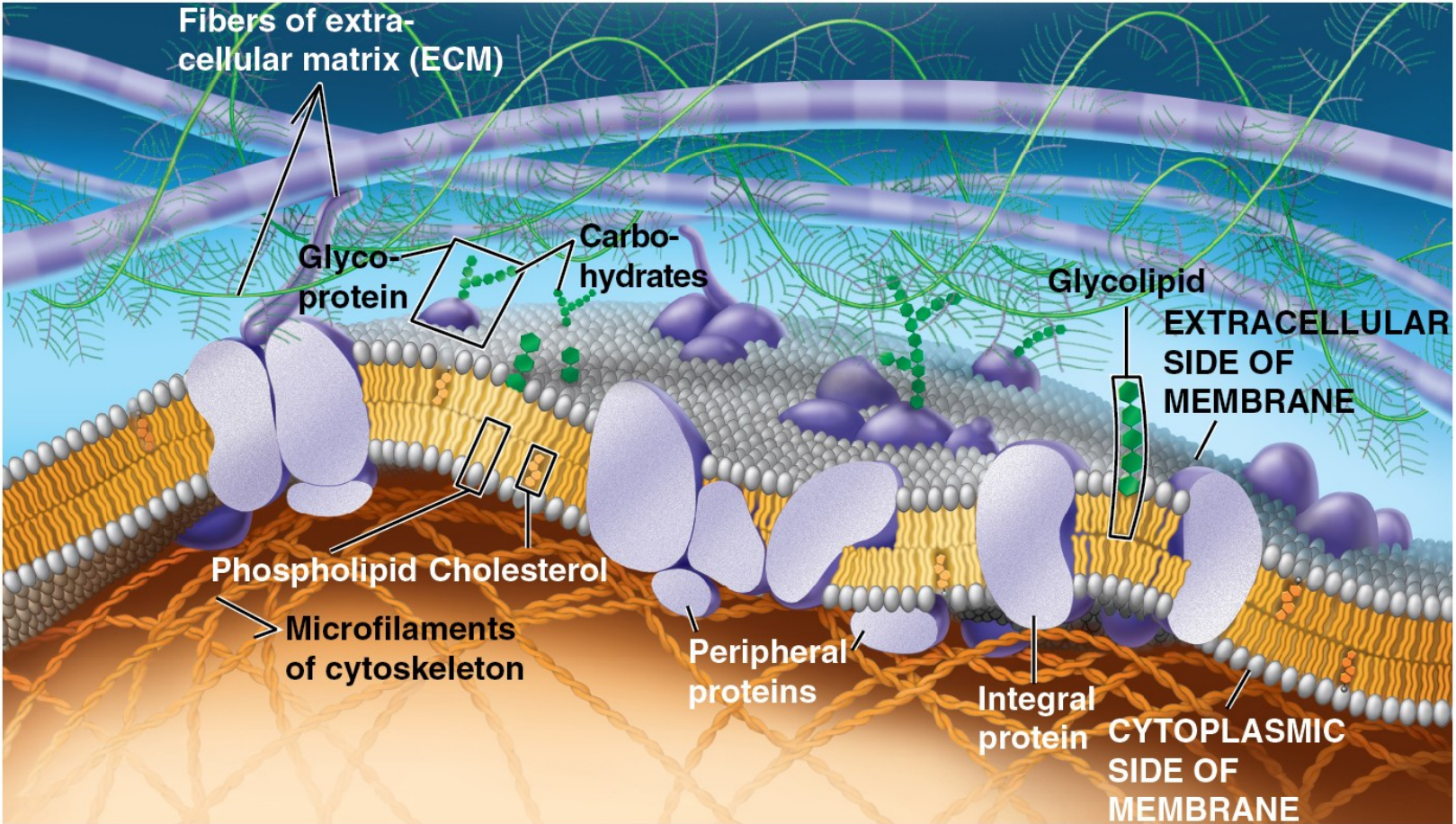
- **Phospholipids** form a bilayer with hydrophobic tails inside the membrane, and hydrophilic heads exposed to water on either side
- Most membrane proteins are also amphipathic
- Hydrophilic regions of the protein are oriented toward the cytosol and extracellular fluid inside and outside the membrane
- Hydrophobic regions are embedded in the bilayer

Figure 8.2



- The **fluid mosaic model** of membrane structure depicts the membrane as a mosaic of protein molecules bobbing in a fluid bilayer of phospholipids
- Proteins are not randomly distributed in the membrane; they often form groups that carry out common functions

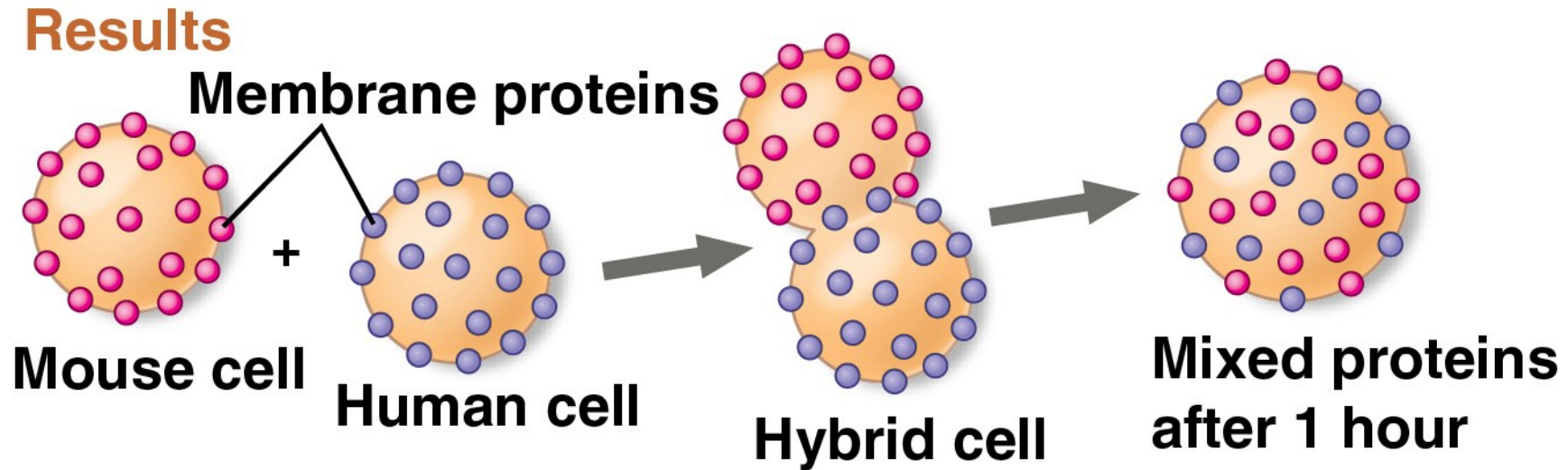
Current model of an animal cell's plasma membrane (cutaway view)



The Fluidity of Membranes

- Membranes are held together mainly by weak hydrophobic interactions
- Most of the lipids and some proteins can move sideways within the membrane
- Rarely, a lipid may flip-flop across the membrane, from one phospholipid layer to the other

Do membrane proteins move?



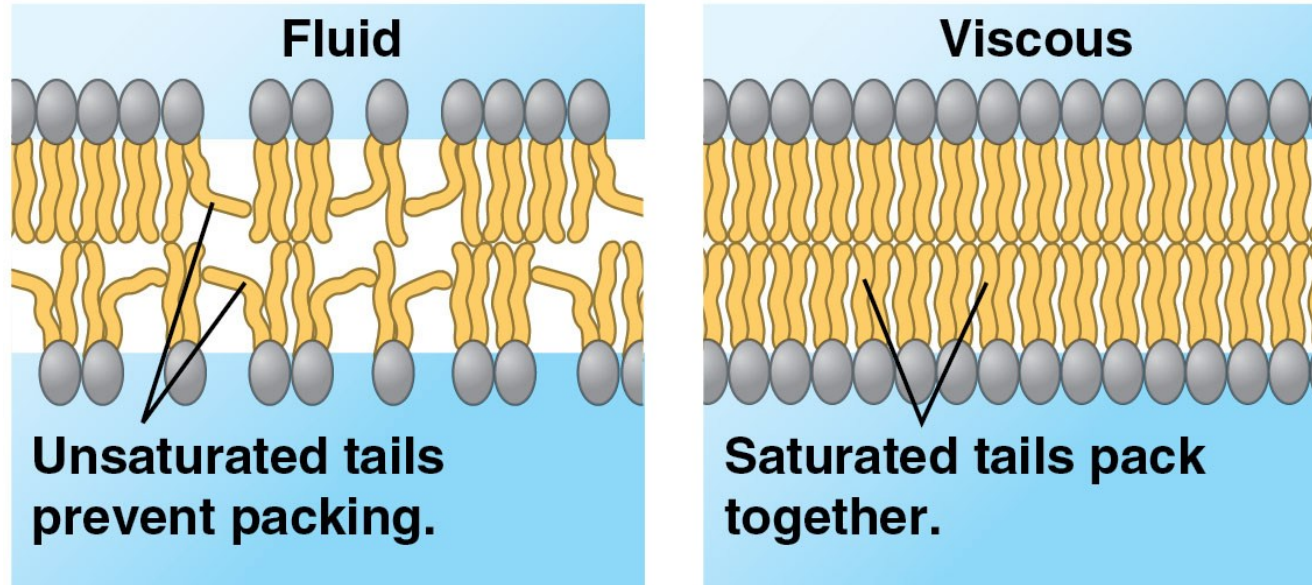
Data from L. D. Frye and M. Edidin, The rapid intermixing of cell surface antigens after formation of mouse-human heterokaryons, *Journal of Cell Science* 7:319 (1970).

- As **temperatures** cool, membranes switch from a fluid state to a solid state
- The temperature at which a membrane solidifies depends on the types of lipids
- Membranes rich in unsaturated fatty acids are more fluid than those rich in saturated fatty acids
- Membranes must be fluid to work properly

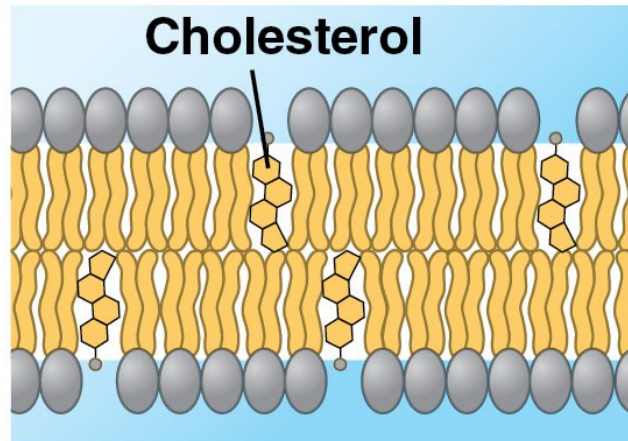
- **Cholesterol** is a membrane component in animal cells that has variable effects on membrane fluidity at different temperatures
 - At warm temperatures (such as 37°C), cholesterol restrains movement of phospholipids
 - At cool temperatures, it maintains fluidity by preventing tight packing
- Plants use different but related steroid lipids to buffer membrane fluidity

Factors that affect membrane fluidity

(a) Unsaturated versus saturated hydrocarbon tails



(b) Cholesterol within the animal cell membrane



Cholesterol reduces membrane fluidity at moderate temperatures by reducing phospholipid movement, but hinders solidification at low temperatures by disrupting packing.

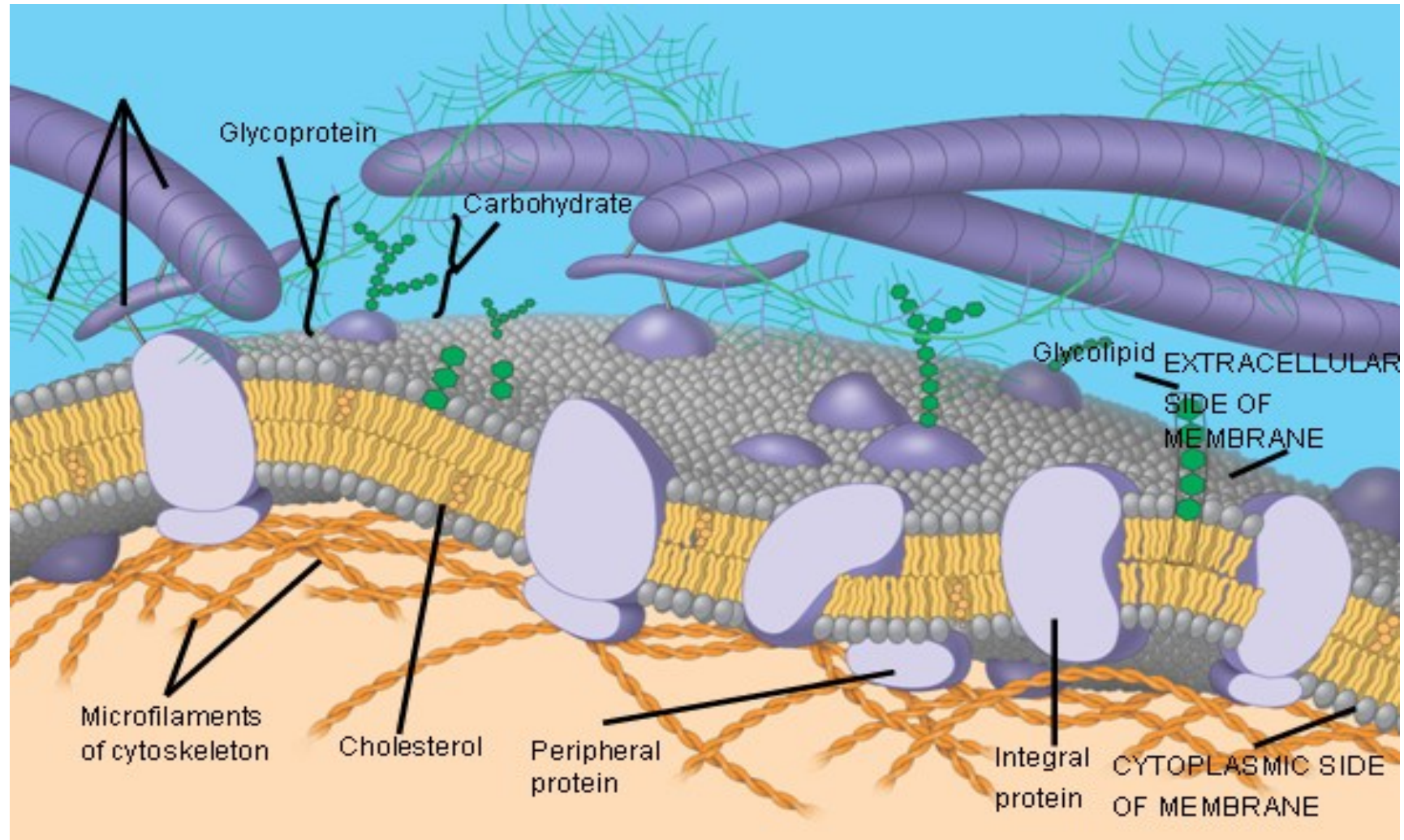
- Membranes must be fluid to work properly; fluidity affects both permeability and movement of transport proteins
- Membranes that are too fluid cannot support protein function
- Organisms living in extreme temperatures have adaptive differences in membrane lipid composition

Evolution of Differences in Membrane Lipid Composition

- The lipid composition of the cell membrane appears to be adapted to environmental conditions in many species
 - For example, cell membranes have a high proportion of unsaturated hydrocarbon tails in fish that live in extreme cold

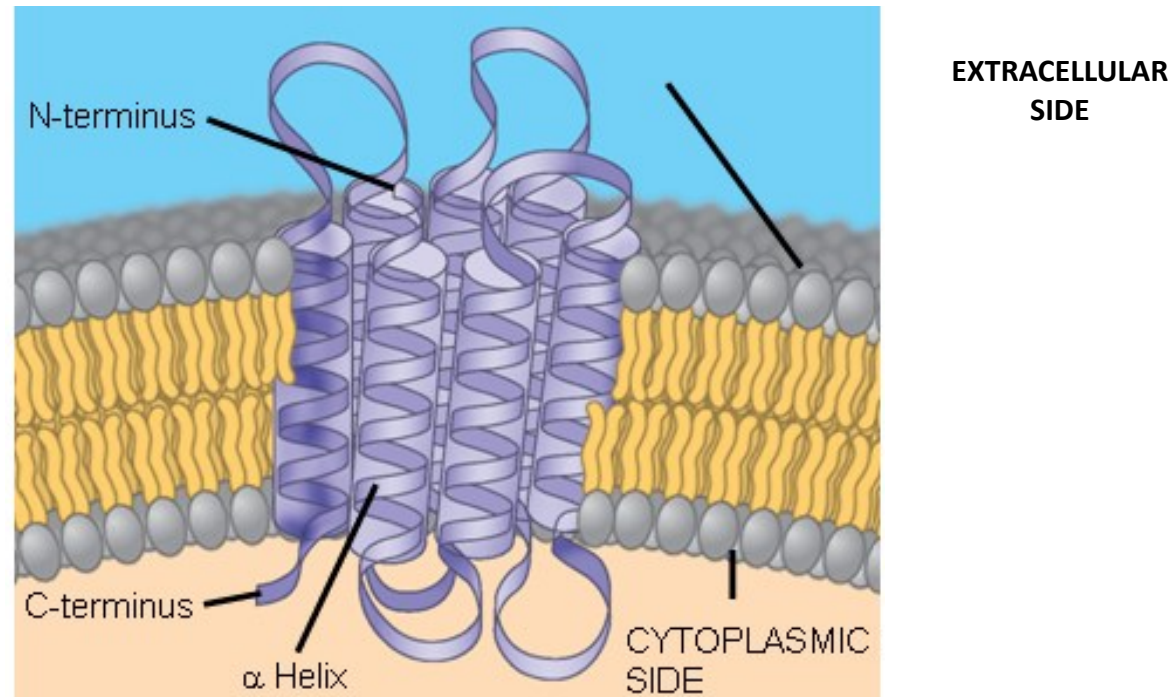
- Organisms living in variable temperature conditions are able to change lipid composition in response to changing temperature
 - For example, in winter wheat, the percentage of unsaturated phospholipids increases in autumn to prevent membrane solidification during winter

Membrane Proteins and Their Functions



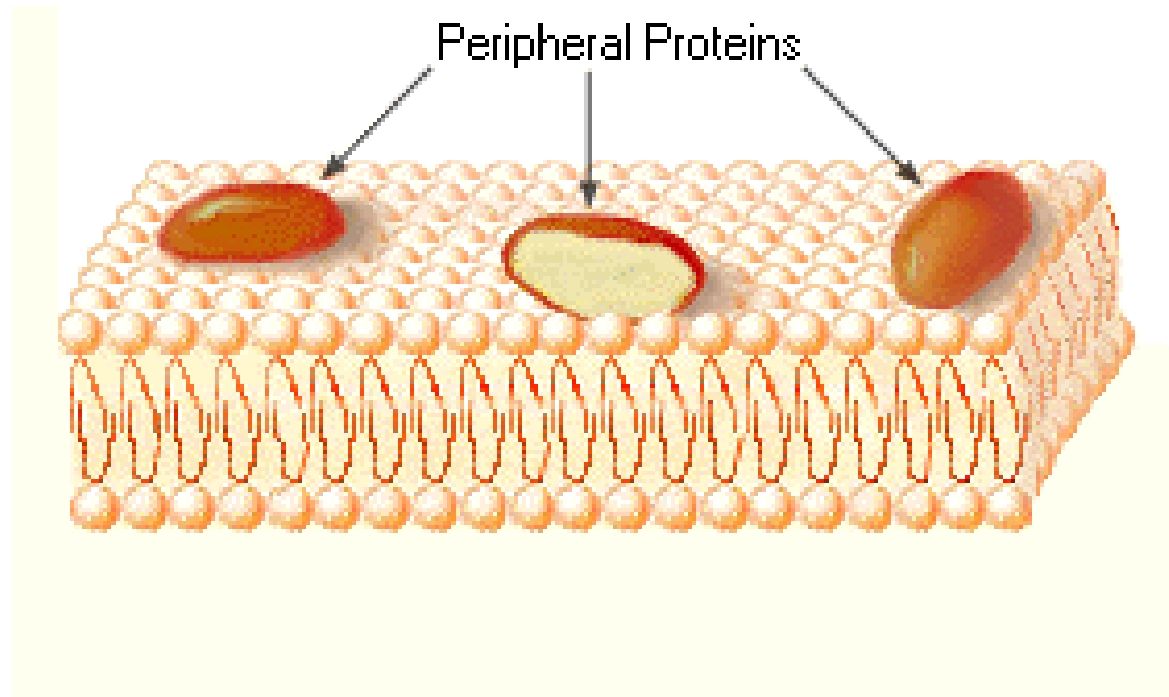
Types of Membrane Proteins

- Integral proteins
 - Penetrate the hydrophobic core of the lipid bilayer
 - Are often transmembrane proteins, completely spanning the membrane



Types of Membrane Proteins

- Peripheral proteins
 - Are appendages loosely bound to the surface of the membrane



- There are two major types of membrane proteins:
 - **Peripheral proteins** are bound to the surface of the membrane
 - **Integral proteins** penetrate the hydrophobic core
- **Transmembrane proteins** are integral proteins that span the membrane
- Hydrophobic regions of an integral protein consist of nonpolar amino acids, often coiled into α helices

Six Major Functions Of Membrane Proteins

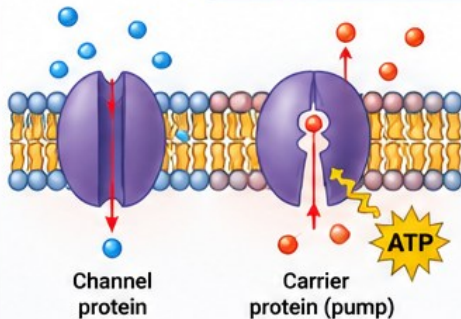
Some Functions of Membrane Proteins

Diverse Roles, Essential for Life



Membrane proteins are embedded in the lipid bilayer and perform a wide range of essential functions that keep cells alive, communicating, and organized.

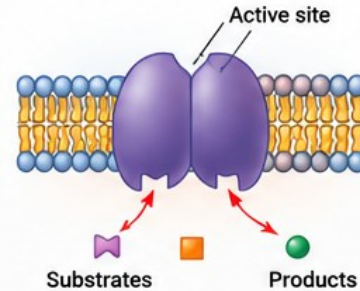
(a) TRANSPORT



- Moves substances across the membrane. Includes channels, carriers, and pumps that transport ions, nutrients, wastes, and other molecules.

Examples: Ion channels, Glucose transporters, Na^+/K^+ pump

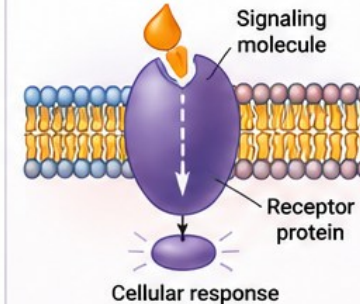
(b) ENZYMATIC ACTIVITY



- Act as enzymes that catalyze specific chemical reactions on the surface of the membrane.

Examples: Digestive enzymes, ATP synthase

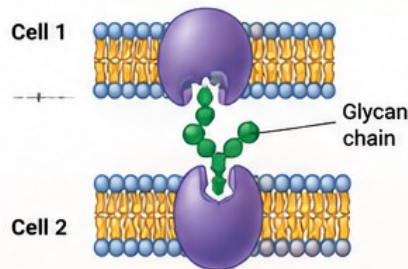
(c) SIGNAL TRANSDUCTION



- Receive chemical signals from outside the cell and trigger responses inside the cell.

Examples: Hormone receptors, Neurotransmitter receptors

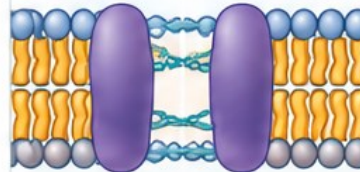
(d) CELL-CELL RECOGNITION



- Enable cells to recognize and identify each other through specific protein markers and carbohydrate chains.

Examples: Immune cell recognition, Cell surface markers

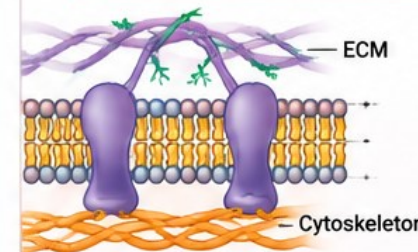
(e) INTERCELLULAR JOINING



- Anchor adjacent cells together, forming tight junctions, gap junctions, and desmosomes.

Examples: Tight junction proteins, Connexins

(f) ATTACHMENT TO CYTOSKELETON AND EXTRACELLULAR MATRIX (ECM)



- Anchor cells to the cytoskeleton and ECM, maintaining cell shape, allowing movement, and transmitting mechanical signals.

Examples: Integrins, Dystrophin



KEY TAKEAWAY

Membrane proteins are multifunctional machines that control what enters and leaves the cell, catalyze reactions, communicate with the environment, and provide structural connections. Their proper function is vital for cell survival and overall organism health.

KEY TERMS



Channel
Forms pores for passive movement



Pump/Carrier
Uses energy to move substances



Enzyme
Catalyzes chemical reactions

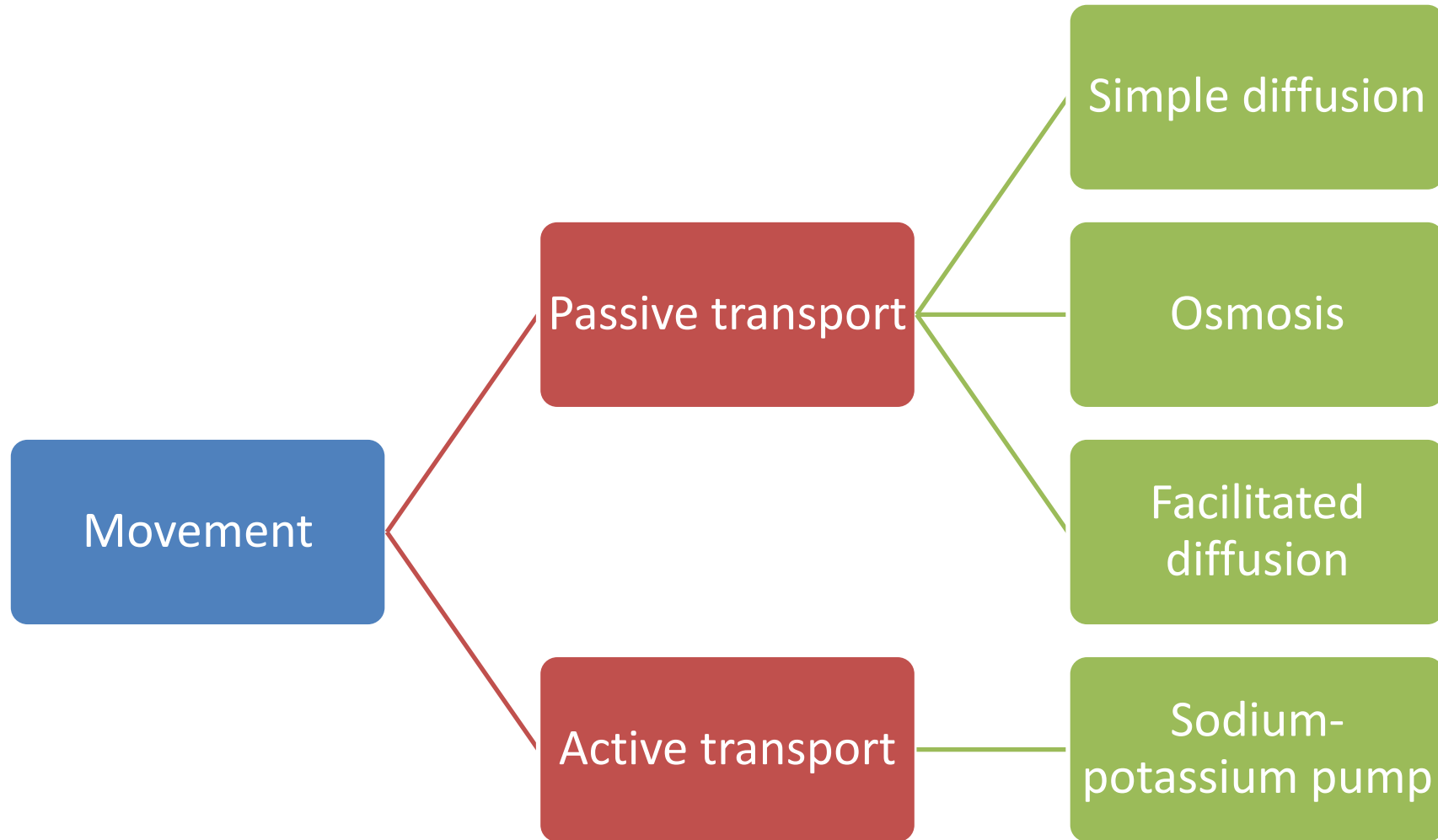


Receptor
Detects signals and triggers response



Linkage
Connects cells or structures

Types of movement across plasma membrane

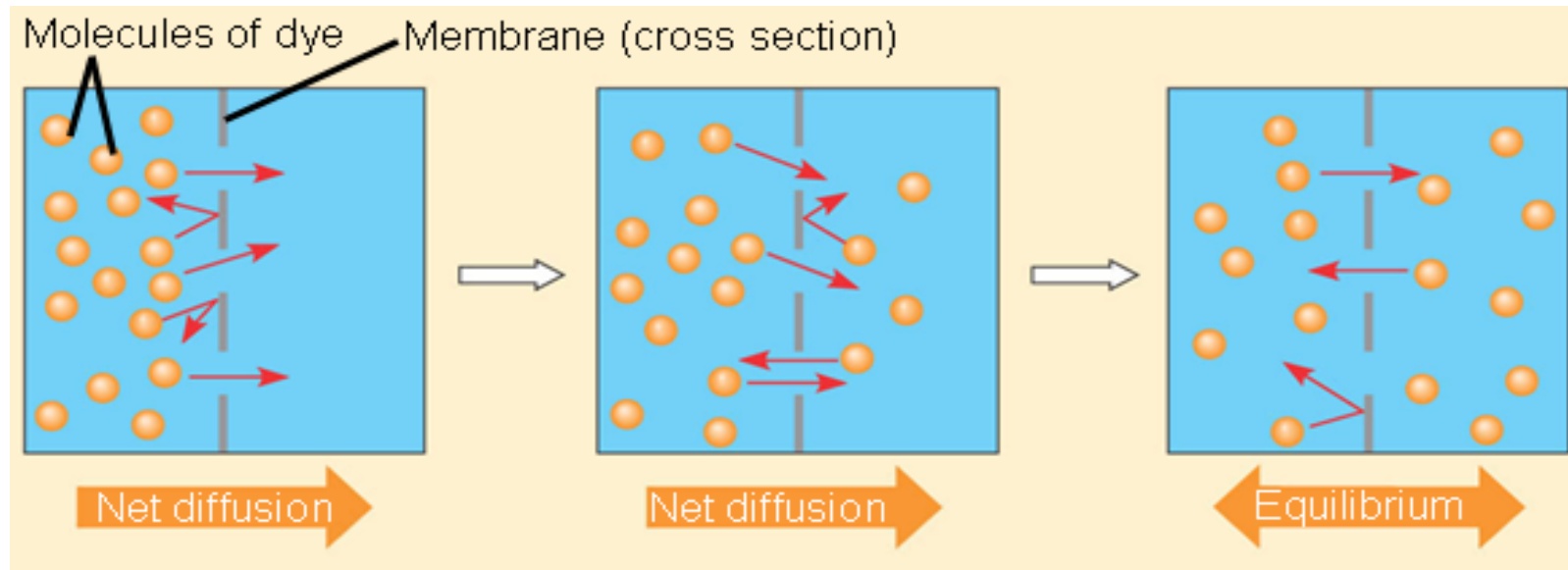


Passive transport

- X need ATP
- Follows the [] gradient
- Net movement of solute stops when reached equilibrium
- May/may not need a transport protein
- If required transport protein then it is specific

Simple Diffusion

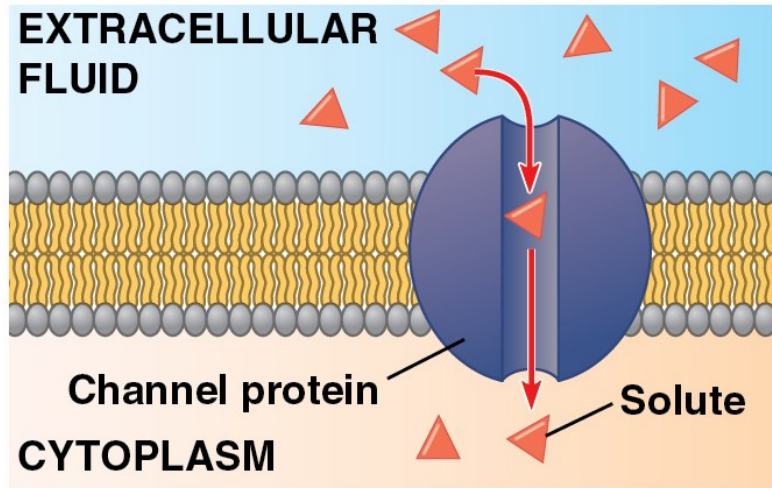
- Diffusion
 - Is the tendency for molecules of any substance to spread out evenly into the available space
 - Move from high to low concentration
 - Down the concentration gradient



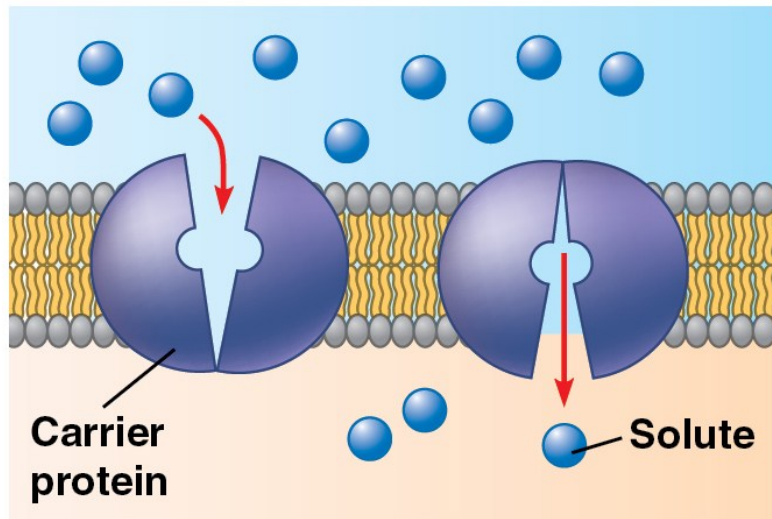
Factors Affecting Diffusion

- I. [] gradient
- II. Temperature
- III. Surface area of membrane
- IV. Size of diffusing molecule
- V. Polarity of diffusing molecule
- VI. Distance over which diffusion occurs

Facilitated Diffusion



(a) A channel protein



(b) A carrier protein

- ✓ H₂O soluble molecules/ molecules x soluble in lipids (ions, nucleic acids, a. a, glucose)

- ✓ special transport proteins: Channel & carrier proteins

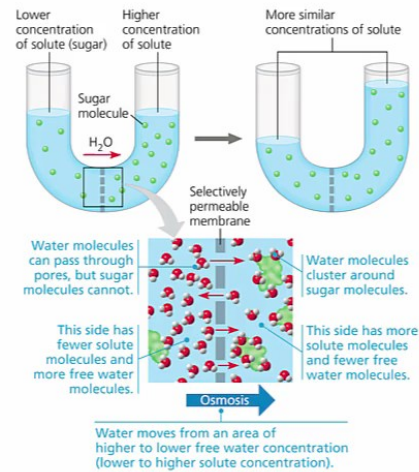
- ✓ down [] gradient , reached equilibrium when [molecules] = both sides of membrane

- ✓ Eg: ions (Na & K in nerve cell without impulse & glucose in/out RBC

Video: Osmosis

CAMPBELL FIGURE WALKTHROUGH

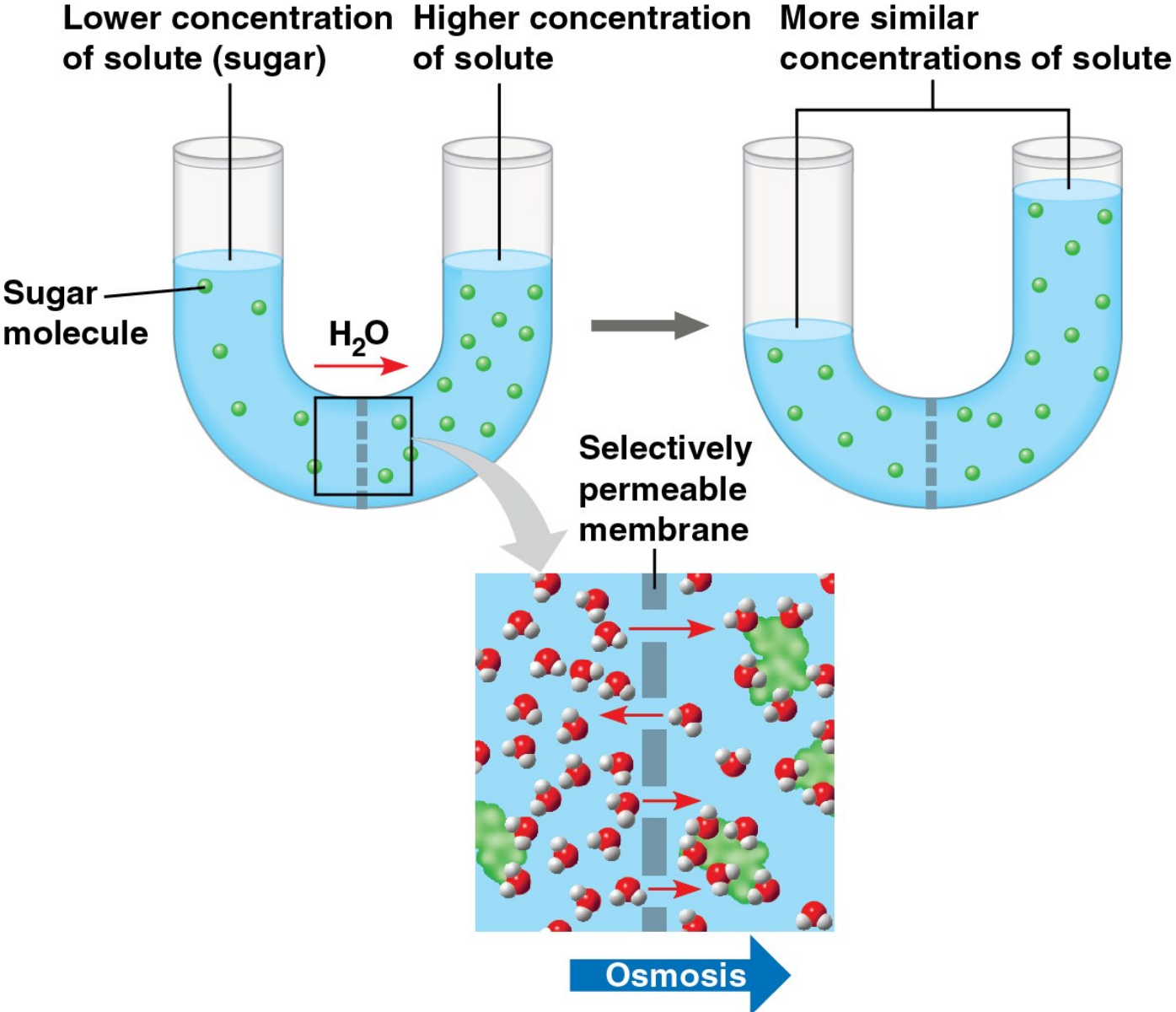
Osmosis



Effects of Osmosis on Water Balance

- **Osmosis** is the diffusion of free water (water molecules not clustered around another substance) across a selectively permeable membrane
- Free water molecules diffuse across a membrane from the region of lower solute concentration to the region of higher solute concentration
- Water keeps moving until the solute concentration is equal on both sides

Figure 8.12



Water Balance of Cells Without Cell Walls

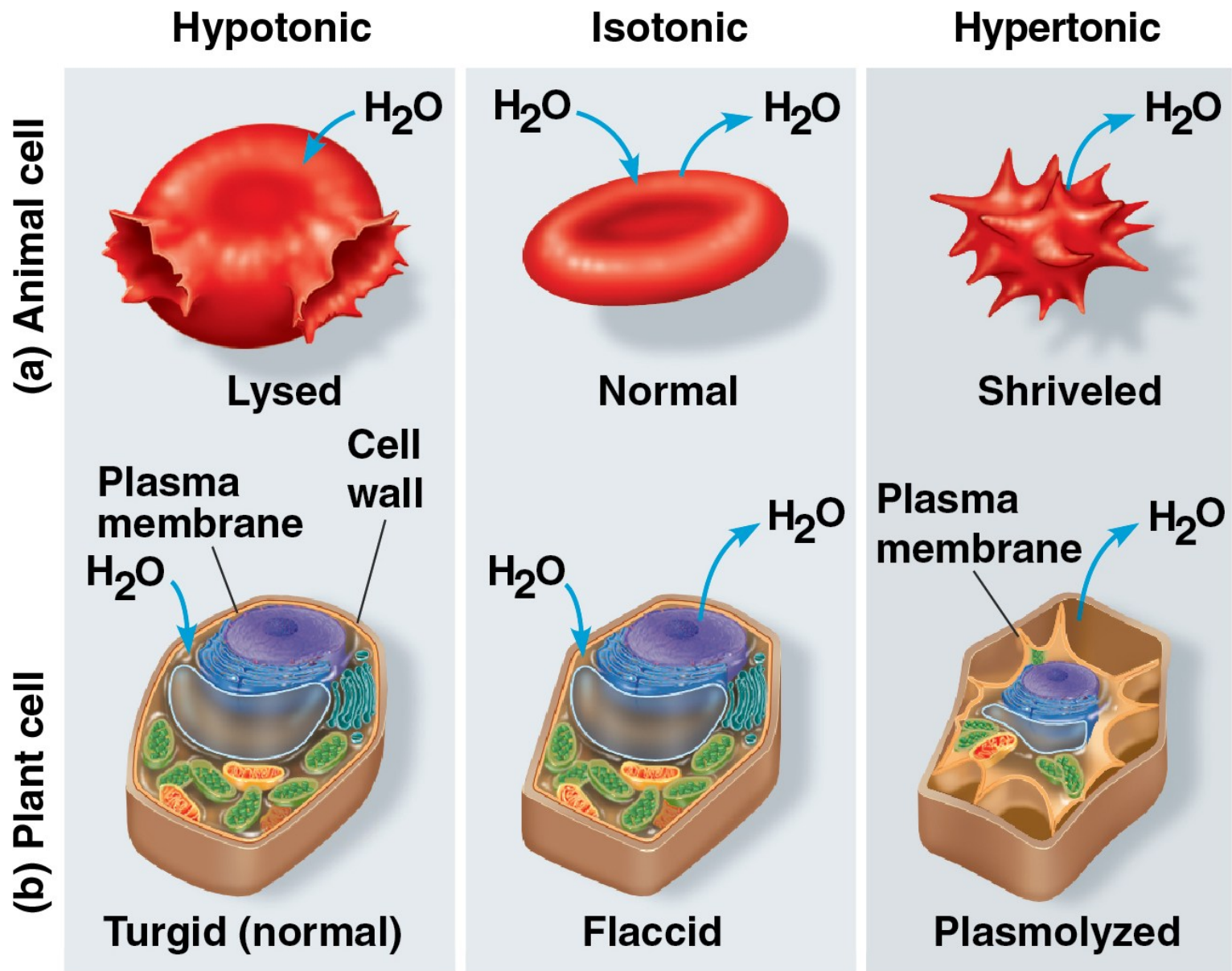
- **Tonicity** is the ability of a surrounding solution to cause a cell to gain or lose water
- Tonicity depends on the concentration of solutes in the solution that cannot cross the membrane, relative to that inside the cell
- If the solution has a higher concentration of these solutes than the inside of the cell, water will tend to leave the cell, and vice versa

- A solution is **isotonic** if its solute concentration is the same as that inside the cell
- Water diffuses across the membrane at the same rate in both directions; there is no *net* movement of water across the membrane
- The volume of a cell without a cell wall is stable in an isotonic solution

- A solution is **hypertonic** if the solute concentration is greater than that inside the cell
- Net diffusion of water is from inside the cell to the surrounding solution
- Cells without cell walls will lose water, shrivel, and likely die in hypertonic solution

- A solution is **hypotonic** if the solute concentration is less than that inside the cell
- Net diffusion of water is from the surrounding solution to the inside of the cell
- Cells without cell walls will gain water, swell and lyse (burst) in a hypotonic solution

Figure 8.13



- Hypotonic or hypertonic environments cause problems for cells without walls because they cannot tolerate excessive water loss or uptake
- Organisms that live in such environments require a method of **osmoregulation**, control of solute concentration and water balance
 - For example, *Paramecium* live in a hypotonic environment; they have a contractile vacuole to pump excess water out of the cell

Water Balance of Cells with Cell Walls

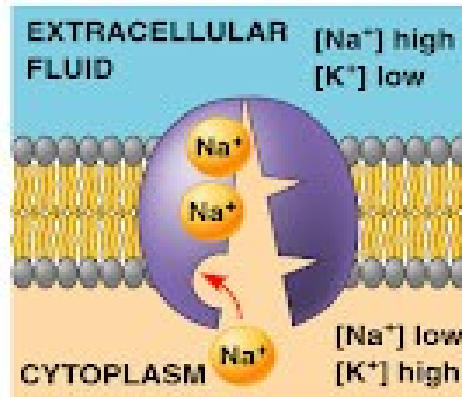
- The cells of plants, prokaryotes, fungi and some protists have cell walls
- A plant cell in a hypotonic solution takes up water and swells until the inelastic wall exerts back a pressure on the cell, called turgor pressure
- At this point the cell is **turgid** (very firm), the healthy state for most plant cells

- If a plant cell and its surroundings are isotonic, there is no net movement of water into the cell
- Plant cells become **flaccid** (limp) in an isotonic solution, and the plant wilts
- Plant cells lose water in a hypertonic environment
- The cell shrivels and the membrane pulls away from the cell wall in multiple locations, a phenomenon called **plasmolysis**
- In response, the plant will wilt and may die

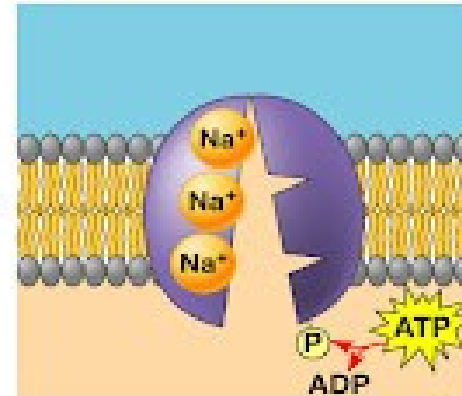
Active Transport

- Transport of substances across membrane against [] gradient
- ATP is needed
- Carrier protein needed
- When there are a lot of substances inside the cell, more can still be taken in
- receptor site (specific)

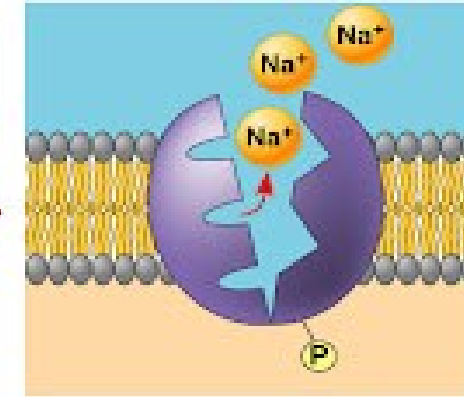
- Factors affecting:
 - Temperature, $[O_2]$, amount of mitochondria, surface area, inhibitors
- Eg:
 - mineral ions intake into root hair cell & young epidermal cells of roots
 - Digested food (glucose) intake into epithelial cells of small intestine with the help of proton pump
 - Sodium-potassium pump maintains potential difference between inside and outside of nerve cells



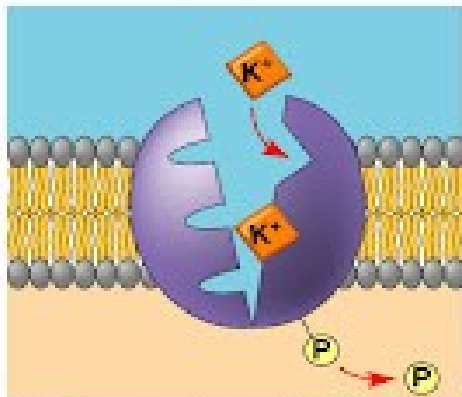
1 Cytoplasmic Na^+ binds to the sodium-potassium pump.



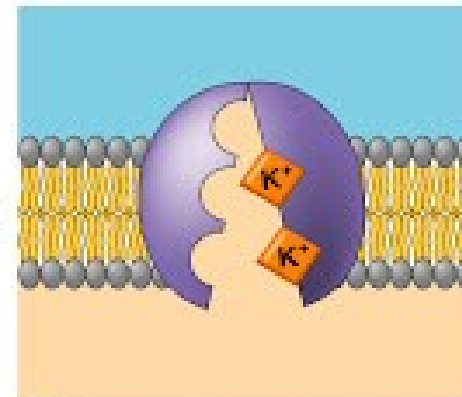
2 Na^+ binding stimulates phosphorylation by ATP.



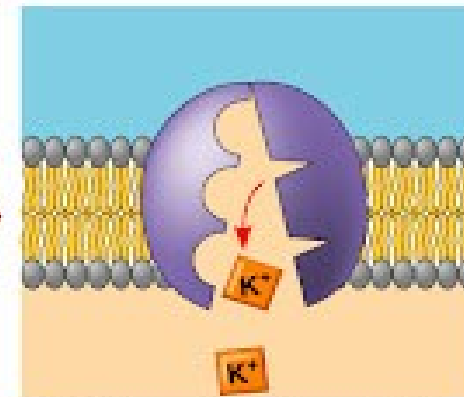
3 Phosphorylation causes the protein to change its conformation, expelling Na^+ to the outside.



4 Extracellular K^+ binds to the protein, triggering release of the phosphate group.

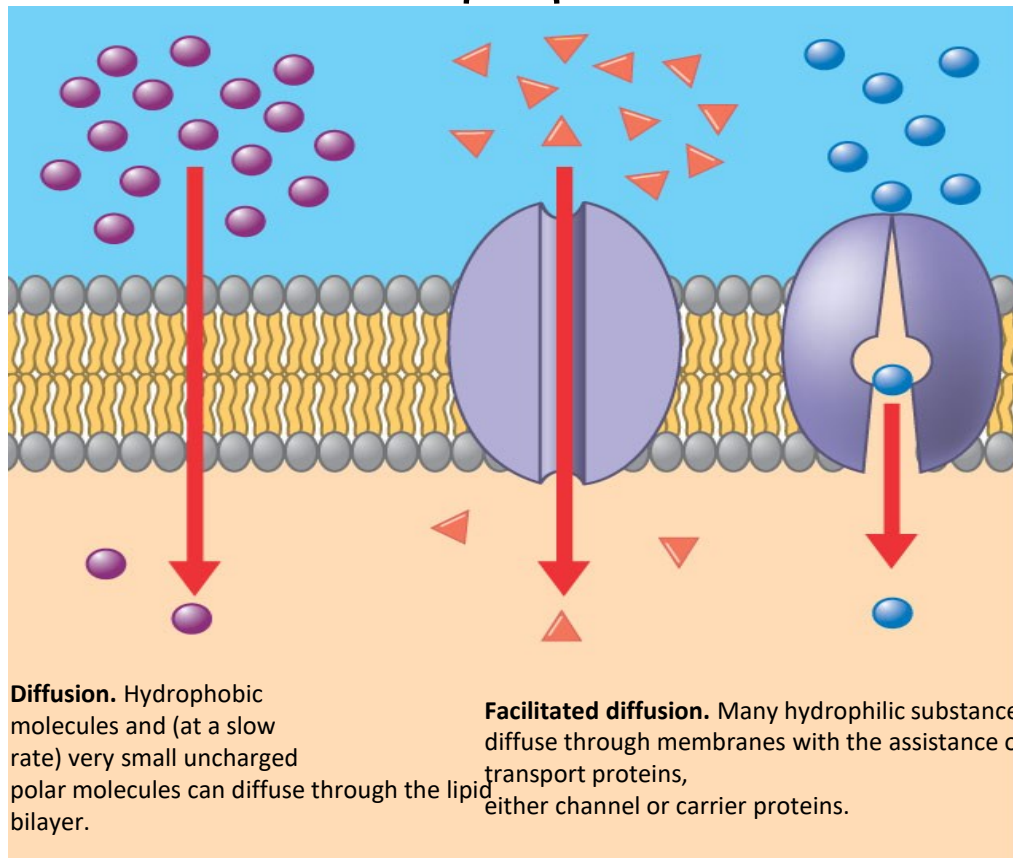


5 Loss of the phosphate restores the protein's original conformation.

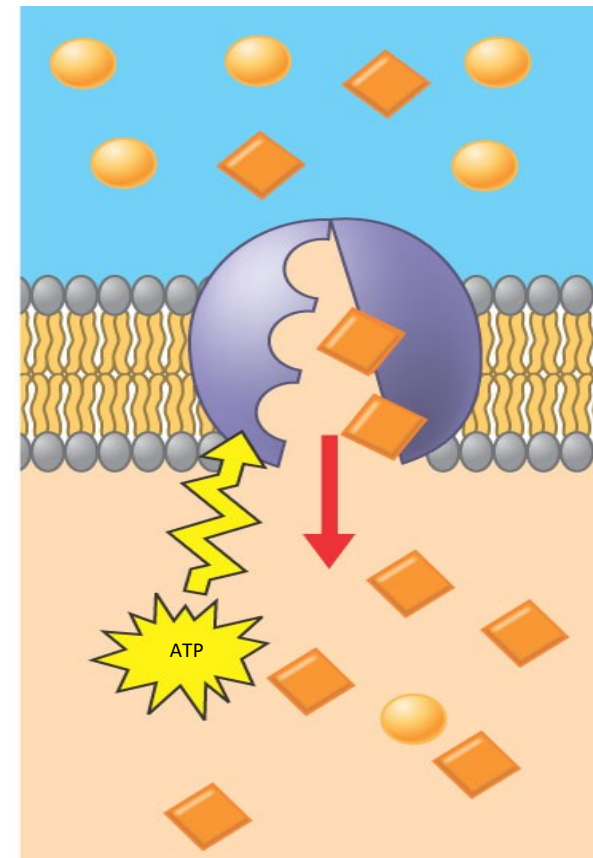


6 K^+ is released and Na^+ sites are receptive again; the cycle repeats.

Passive transport. Substances diffuse spontaneously down their concentration gradients, crossing a membrane with no expenditure of energy by the cell. The rate of diffusion can be greatly increased by transport proteins in the membrane.



Active transport. Some transport proteins act as pumps, moving substances across a membrane against their concentration gradients. Energy for this work is usually supplied by ATP.



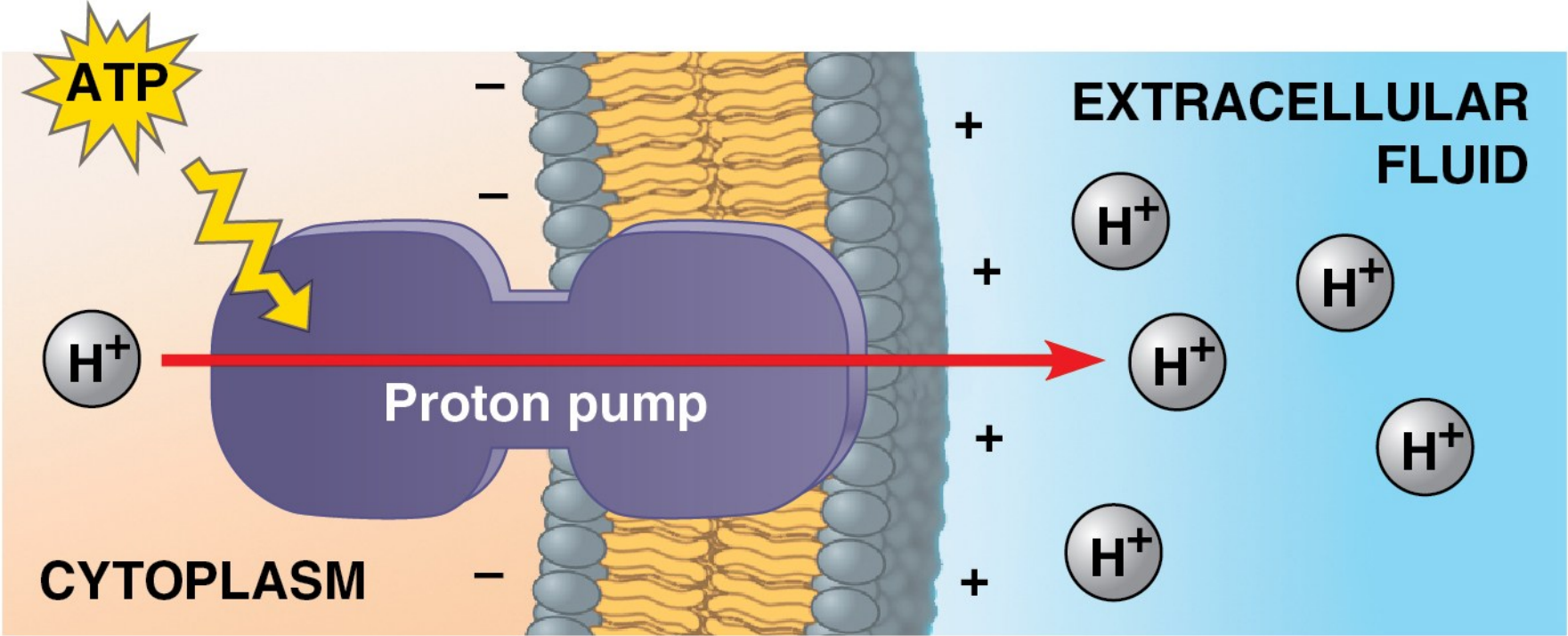
How Ion Pumps Maintain Membrane Potential

- **Membrane potential** is the voltage across a membrane
- Voltage is created by differences in the distribution of positive and negative ions across a membrane
- The inside of the cell is negative in charge relative to the outside, favoring passive transport of cations into and anions out of the cell

- Two combined forces, collectively called the **electrochemical gradient**, drive the diffusion of ions across a membrane
 - A chemical force (the ion's concentration gradient)
 - An electrical force (the effect of the membrane potential on the ion's movement)
- An ion diffuses down its electrochemical gradient

- An **electrogenic pump** is a transport protein that generates voltage across a membrane, storing energy that can be used for cellular work
- The main electrogenic pump differs between plants and animals
 - In animals, it is the sodium-potassium pump
 - In plants, fungi, and bacteria, it is the **proton pump**, which actively transports hydrogen ions (H^+) out of the cell

Figure 8.18



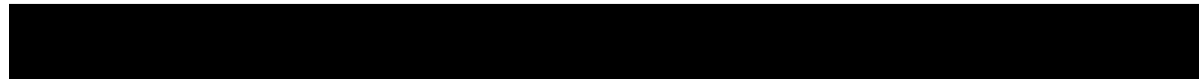
Bulk Transport Across Plasma Membrane

- Large molecules : proteins & polysaccharides across plasma membrane in bulk via packaging in vesicles and ATP is needed
- Endocytosis (Uptake of substances)
- Exocytosis (Egests/secretates substances)

Animation: Exocytosis and Endocytosis

Introduction

Exocytosis and Endocytosis



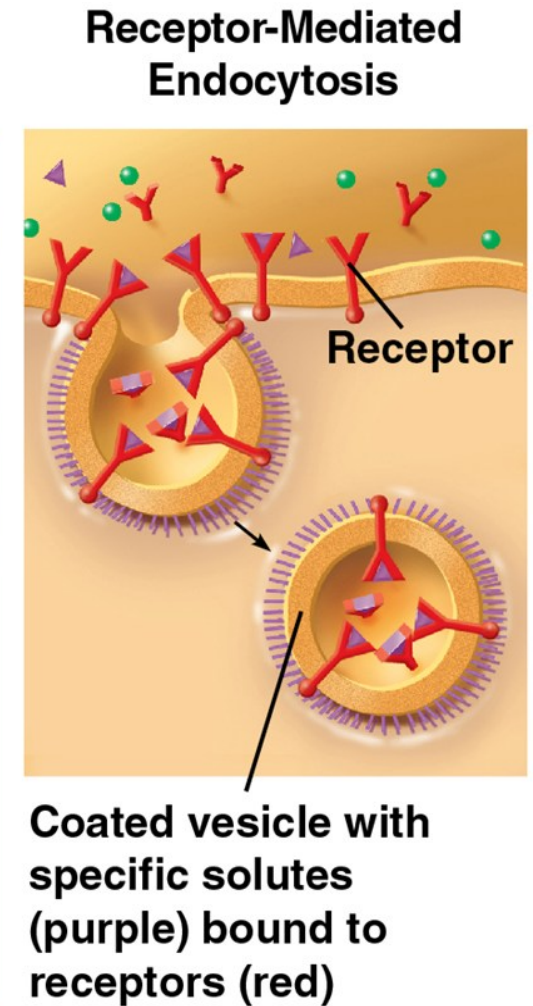
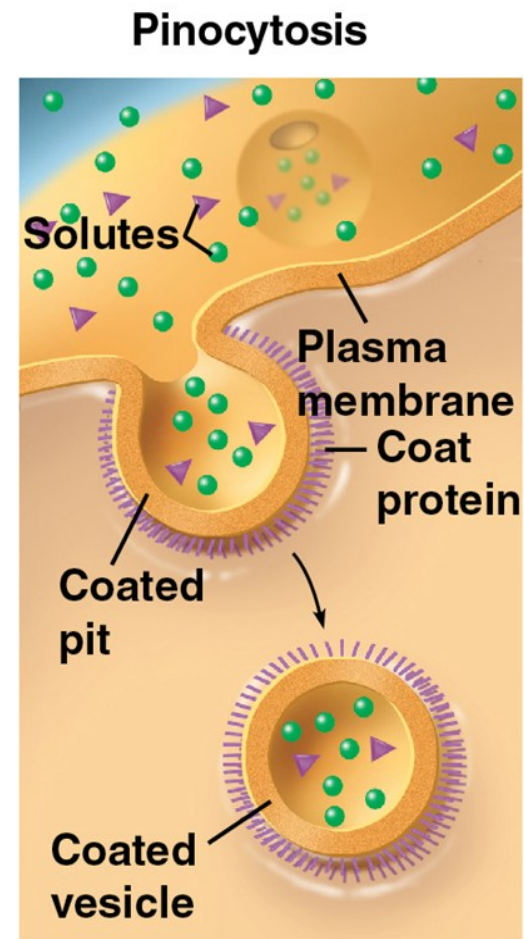
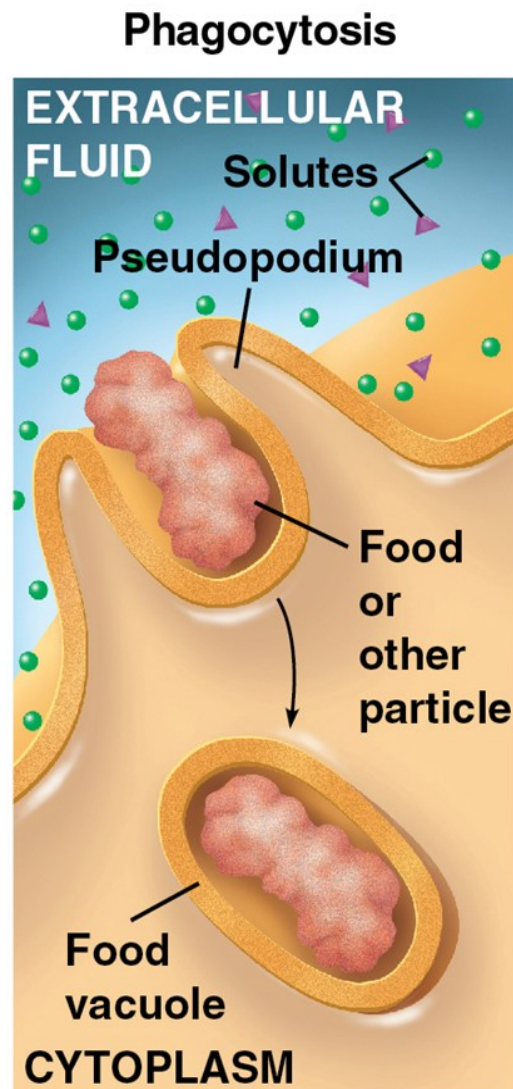
Exocytosis

- In **exocytosis**, transport vesicles migrate to the membrane, fuse with it, and release their contents outside the cell
- Many secretory cells use exocytosis to export their products
 - For example, cells in the pancreas secrete insulin by exocytosis

Endocytosis

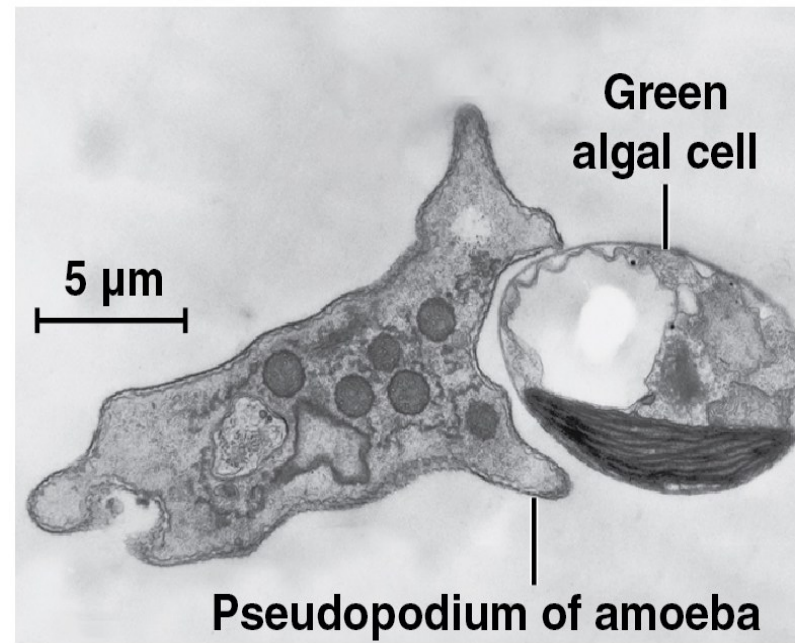
- In **endocytosis**, macromolecules are taken into the cell in vesicles
- The membrane forms a pocket that deepens and pinches off forming a vesicle around the material for transport

- There are three types of endocytosis
 - Phagocytosis (“cellular eating”)
 - Pinocytosis (“cellular drinking”)
 - Receptor-mediated endocytosis

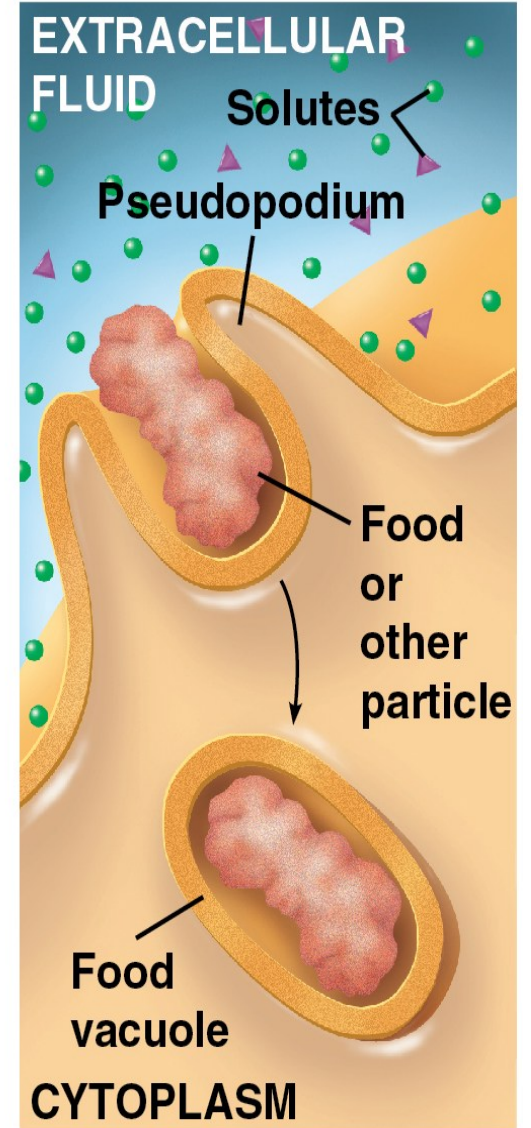


- In **phagocytosis**, a cell engulfs a particle by extending pseudopodia around it and packing it in a membranous sac called a food vacuole
- The vacuole fuses with a lysosome to digest the particle

Phagocytosis

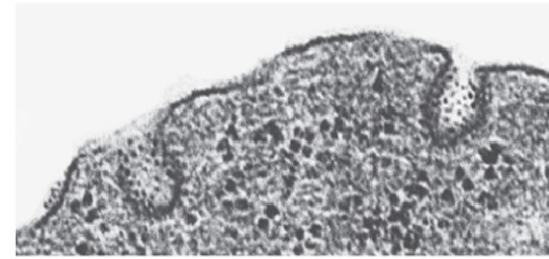
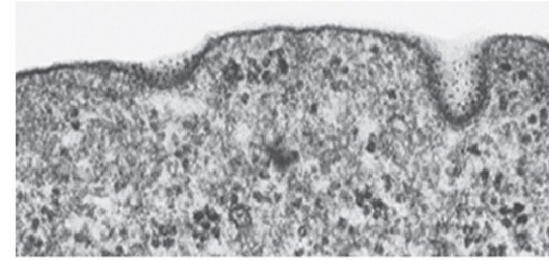


An amoeba engulfing a green algal cell via phagocytosis (TEM)



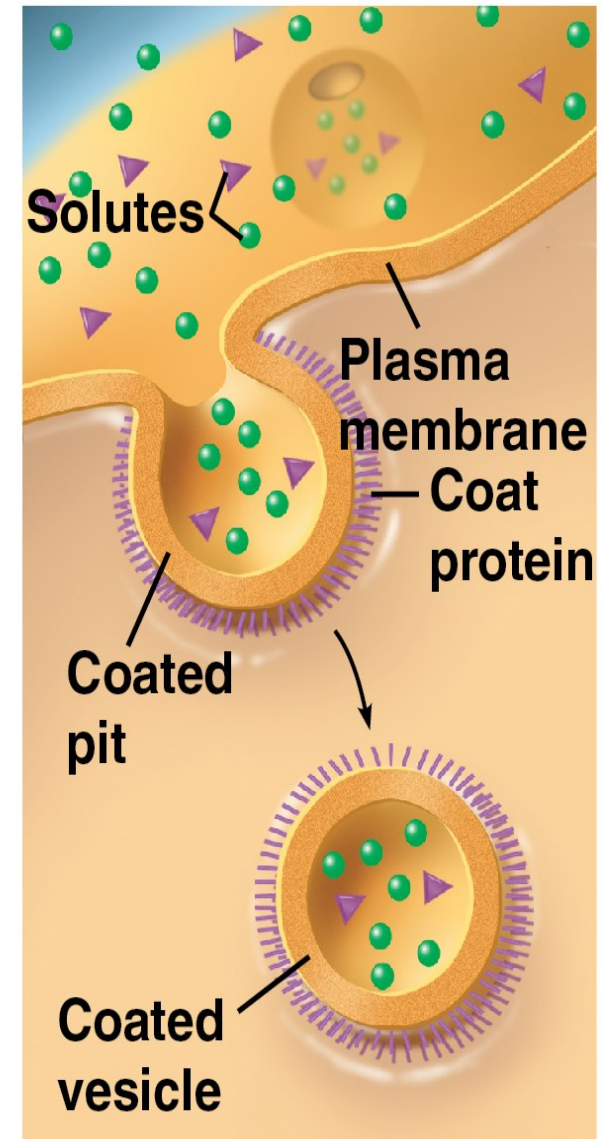
- In **pinocytosis**, molecules are taken up when extracellular fluid is “gulped” into tiny vesicles
- Pinocytosis is nonspecific for the substances it transports; any and all solutes are taken into the cell
- Parts of the plasma membrane that form vesicles are lined on the inner side with coat proteins, forming coated vesicles

Pinocytosis



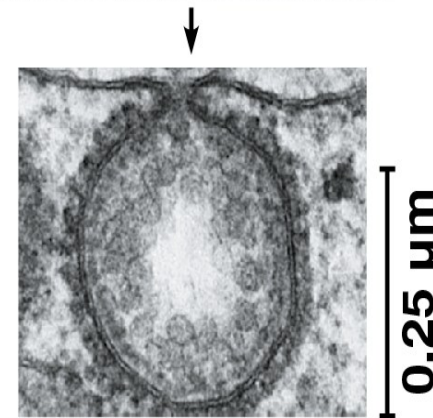
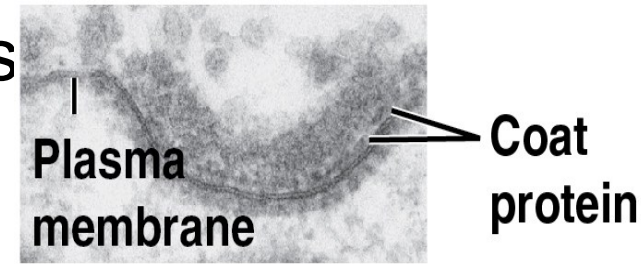
Pinocytotic vesicles forming (TEMs)

0.25 μm

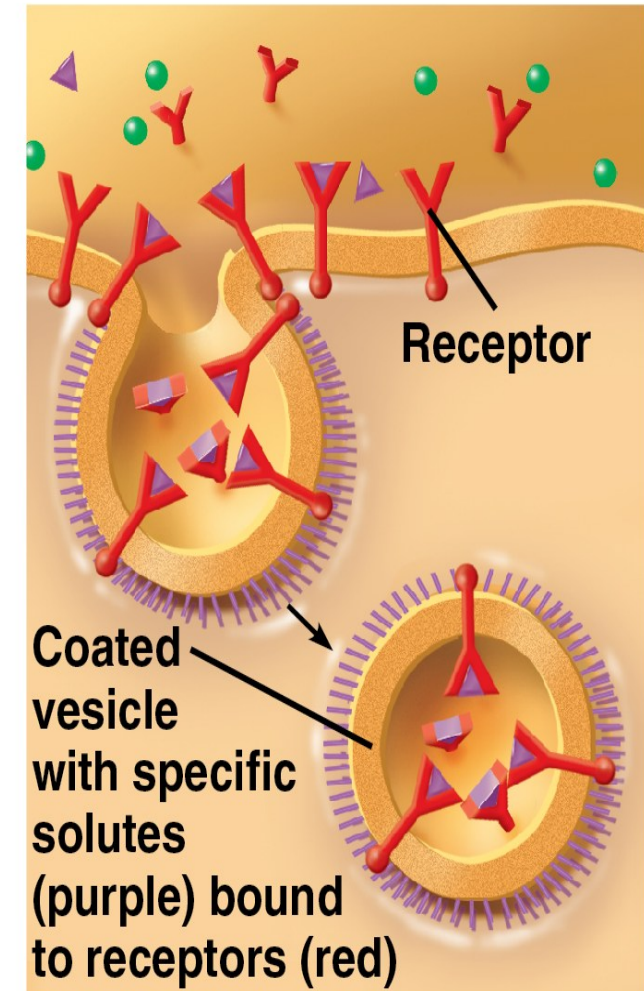


- In **receptor-mediated endocytosis**, vesicle formation is triggered by solute binding to receptors
- Receptor proteins bound to specific solutes from the extracellular fluid are clustered in coated pits that form coated vesicles
- Emptied receptors are recycled to the plasma membrane by the same vesicle

Receptor-Mediated Endocytosis



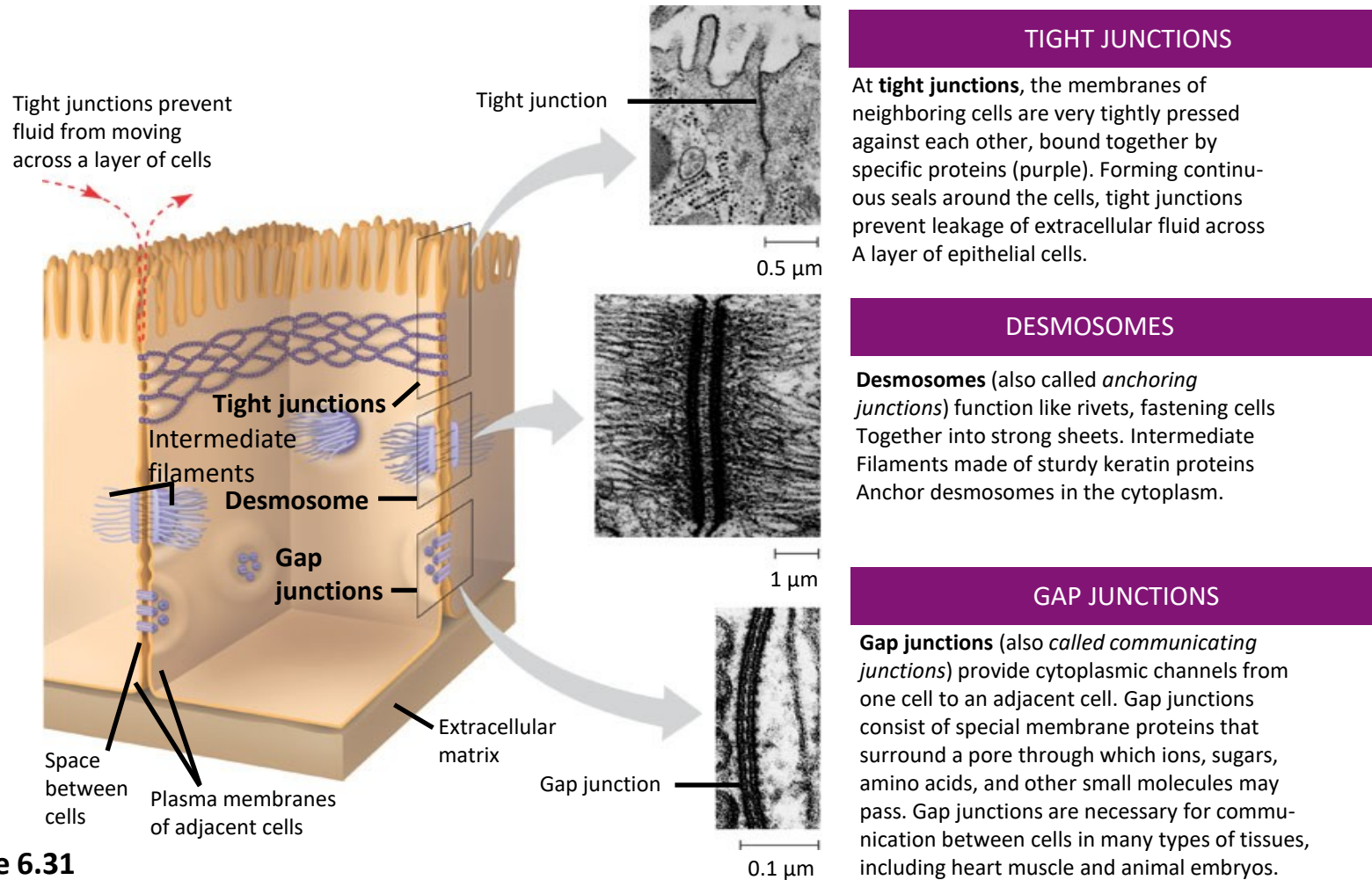
Top: A coated pit
Bottom: A coated vesicle forming during receptor-mediated endocytosis (TEMs)



Cell Junctions

- In animals, there are three types of intercellular junctions
 - Tight junctions
 - Desmosomes
 - Gap junctions

- Types of intercellular junctions in animals

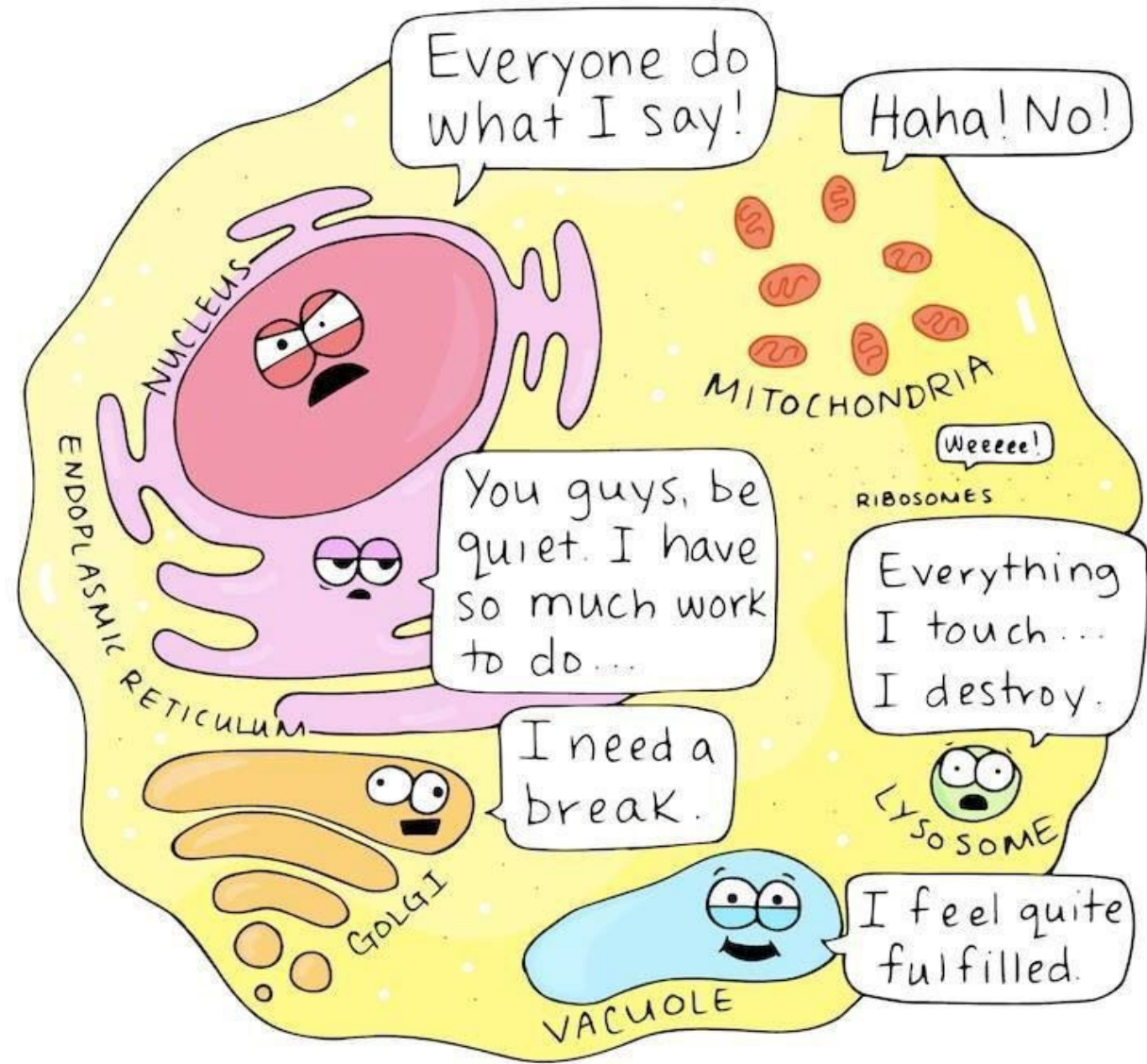


Synthesize your knowledge!



- In the supermarket, lettuce and other produce are often sprayed by water. Explain why this makes vegetables crisp??

Thank you



If organelles could talk.

Beatrice the Biologist